Journal of Volcanology and Geothermal Research 261 (2013) 330–347

Contents lists available at SciVerse ScienceDirect

Journal of Volcanology and Geothermal Research

journal homepage: www.elsevier.com/locate/jvolgeores

Rain-triggered lahars following the 2010 eruption of Merapi volcano, Indonesia: A major risk

Edouard de Bélizal [a,*], Franck Lavigne [a], Danang Sri Hadmoko [b], Jean-Philippe Degeai [a], Gilang Aria Dipayana [b], Bachtiar Wahyu Mutaqin [b], Muh Aris Marfai [b], Marie Coquet [a], Baptiste Le Mauff [a], Anne-Kyria Robin [a], Céline Vidal [c], Noer Cholik [d], Nurnaning Aisyah [d]

[a] University Paris 1 Panthéon Sorbonne and University Paris-Est Créteil (UPEC), Laboratoire de Géographie Physique, CNRS UMR 8591, 1 place A. Briand, 92195 Meudon cedex, France
[b] Center for Natural Disaster Studies (Pusat Studi Bencana Alam PSBA), Gadjah Mada University, Faculty of Geography, Bulaksumur, Yogyakarta, Indonesia
[c] Institut de Physique du Globe de Paris, CNRS UMR 7154, Équipe de Géologie des Systèmes Volcaniques, 4 place Jussieu, 75252 Paris Cedex 05, France
[d] BPPTK (Balai Penyeledikan dan Pengembangan Teknologi Kegunungapian), Jalan Cendana 15, Yogyakarta 55166, Indonesia

ARTICLE INFO

Article history:
Received 7 August 2012
Accepted 21 January 2013
Available online 28 January 2013

Keywords:
Rain-triggered lahars
Lahar corridors
Lahar deposits
Crisis management
Merapi volcano

ABSTRACT

The 2010 VEI 4 eruption of Merapi volcano deposited roughly ten times the volume of pyroclastic materials of the 1994 and 2006 eruptions, and is recognized as one of the most intense eruption since 1872. However, as the eruptive phase is now over, another threat endangers local communities: rain-triggered lahars. Previous papers on lahars at Merapi presented lahar-related risk following small-scale dome-collapse PDCs. Thus the aim of this study is to provide new insights on lahar-related risk following a large scale VEI 4 eruption. The paper highlights the high number of events (240) during the 2010–2011 rainy season (October 2010–May 2011). The frequency of the 2010–2011 lahars is also the most important ever recorded at Merapi. Lahars occurred in almost all drainages located under the active cone, with runout distances exceeding 15 km. The geomorphic impacts of lahars on the distal slope of the volcano are then explained as they directly threaten houses and infrastructures: creation of large corridors, avulsions, riverbank erosion and riverbed downcutting are detailed through local scale examples. Related damage is also studied: 860 houses damaged, 14 sabo-dams and 21 bridges destroyed. Sedimentological characteristics of volcaniclastic sediments in lahar corridors are presented, with emphasis on the resource in building material that they represent for local communities. Risk studies should not forget that thousands of people are exposing themselves to lahar hazard when they quarry volcaniclastic sediment on lahar corridors. Finally, the efficient community-based crisis management is explained, and shows how local people organize themselves to manage the risk: 3 fatalities were reported, although lahars reached densely populated areas. To summarize, this study provides an update of lahar risk issues at Merapi, with emphasis on the distal slope of the volcano where lahars had not occurred for 40 years, and where lahar corridors were rapidly formed.

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

The Indonesian word "lahar" is applied as a general term for rapidly flowing, highly concentrated and poorly-sorted sediment-laden mixtures of water and rock debris from a volcano, not including normal streamflow (Smith and Fritz, 1989; Vallance, 2000). Lahars have been defined as one of the most important hazard at Merapi volcano (Lavigne et al., 2000a,b; Thouret et al., 2000; Lavigne and Thouret, 2002), following the dome-collapse pyroclastic density currents (PDC) which used to occur every 4–6 years during the 20th century until 2006 (Abdurachman et al., 2000; Newhall et al., 2000; Voight et al., 2000; Charbonnier and Gertisser, 2008). Generally, lahars at Merapi volcano are brief events, related to rainstorms which commonly last 1 or 2 h (Lavigne et al., 2000a,b; Lavigne and Thouret, 2002). Since the introduction of sabo-dam structures on the river channels from the late 1970s, it has been possible to slow lahars (Lavigne and Thouret, 2002). Lahars were therefore constrained on the upper part of the rivers, and seldom exceeded a length of 10 km from the crater. As a result, lahar-related damages and casualties at Merapi have been limited since the 1980s (Lavigne et al., 2000a), and mainly occurred at the bottom of the valleys in quarries mining volcaniclastic deposits: 187 trucks were swept away by lahars between 1987 and 2010 (De Bélizal et al., 2011) and no human casualties were reported. The last lahar-related risk assessment at Merapi was made at the end of the 1990s (Lavigne, 1999; Lavigne et al., 2000a) and could be applied mainly to rain-triggered lahars following dome-collapse PDCs.

* Corresponding author.
E-mail addresses: edouard.dbelizal@gmail.com (E. de Bélizal), franck.lavigne@univ-paris1.fr (F. Lavigne), hadmokoo@yahoo.com (D.S. Hadmoko), degeai@cnrs-bellevue.fr (J.-P. Degeai), aryadipayana@gmail.com (G.A. Dipayana), bachtiarwahyumutaqin@gmail.com (B.W. Mutaqin), arismarfai@yahoo.com (M.A. Marfai), coquet.marie@hotmail.fr (M. Coquet), batlm@hotmail.fr (B.L. Mauff), annekyria.robin@free.fr (A.-K. Robin), noer.cholik@gmail.com (N. Cholik), aisy_bpptk@yahoo.com (N. Aisyah).

0377-0273/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.jvolgeores.2013.01.010

The 2010 VEI 4 explosive eruption of Merapi volcano reached a magnitude and intensity larger than the frequent eruptions of the 20th century. About 0.03 to 0.06 km^3 of pyroclastic materials from PDCs and tephra fallout were ejected during the eruption (Thierry et al., 2011; Surono et al., 2012; Komorowski et al., 2013; Charbonnier et al., 2013). This is ten times higher than other Merapi dome-collapse block-and-ash deposits produced in the 20th century (Andreastuti et al., 2000; Newhall et al., 2000; Schwarzkopf et al., 2005; Charbonnier and Gertisser, 2008). Every watershed located under the active cone of the volcano was covered by the 2010 pyroclastic deposits, which raises the issue of the volcaniclastic remobilization of those deposits by rainfalls. Rain-triggered lahars following explosive eruptions can generate long-term risk for people living along the river channels, as the landscape response to the volcanic disturbance can take many years. At Mount Pinatubo (Philippines), the post-1991 eruption lahars occurred during about a decade. They resulted in the loss of more lives than those directly lost from the eruption, and left

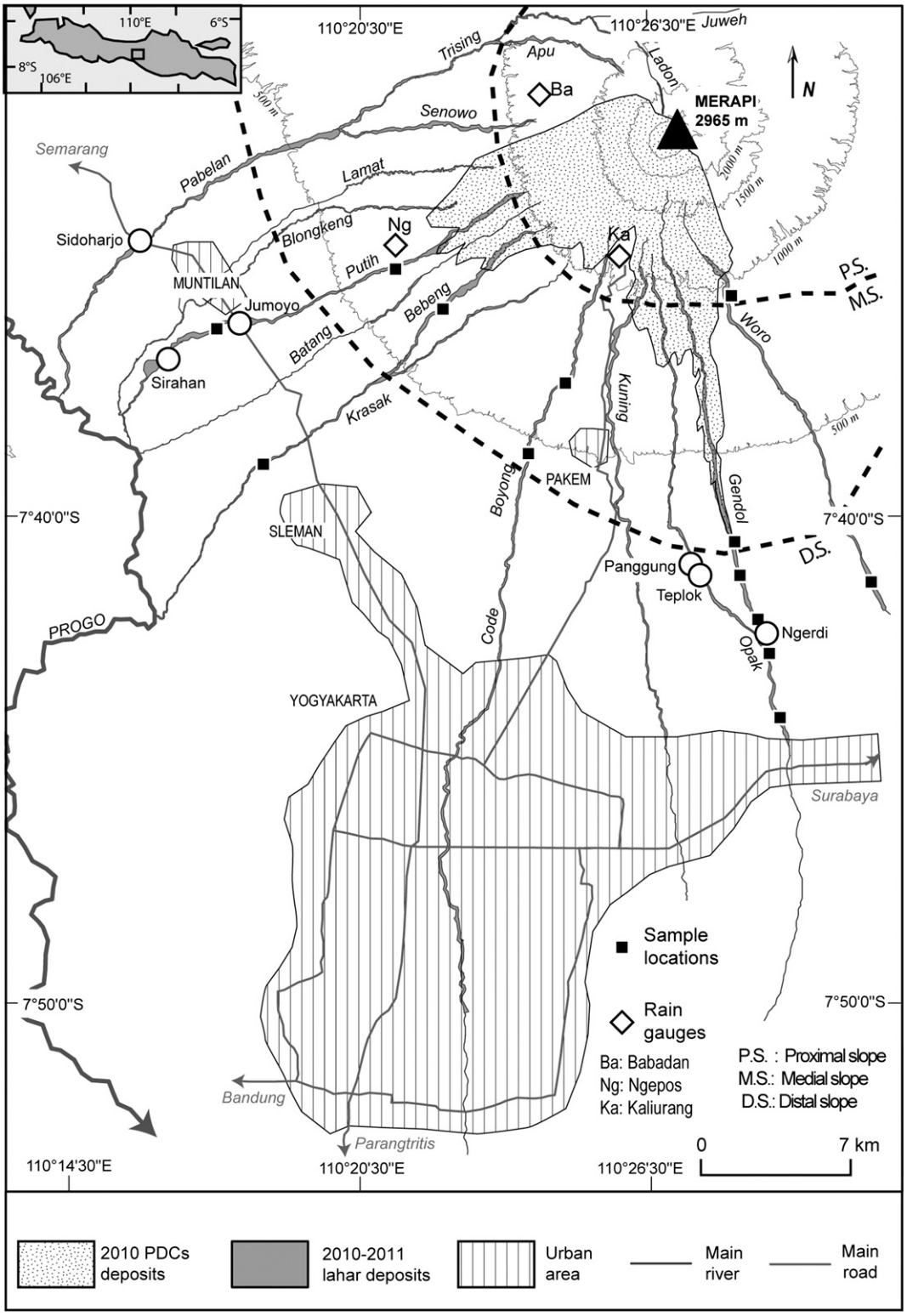


Fig. 1. Location map.

volcaniclastic deposits in excess of 2 km^3 on about 1000 km^2 of the flanks and aprons of the volcano, affecting more than 200,000 people (Bautista, 1996; Major et al., 1996; Pierson et al., 1996; Scott et al., 1996; Gaillard et al., 2001).

Lahar-related risk issue at Merapi volcano thus needs to be updated after the 2010 VEI 4 eruption. This paper aims to define the way lahars have become a major risk at Merapi, and will focus on four issues. (1) The hazard has become more frequent, more widespread and with larger runout distances than what has been shown in previous studies. (2) The longer length of lahars led to geomorphic impacts on river channels on the distal slope of the volcano, creating large corridors which damaged dams, bridges, roads and settlements located along the rivers. (3) The sedimentary materials brought by lahars represent a resource in boulders and sand, attracting hundreds of workers everyday on lahar-prone areas. (4) The self-evacuations and the community-based early warning system which have been developed by local communities in order to prevent themselves against lahars.

2. Methods

This comprehensive study of all the different aspects of lahar-related risks at Merapi volcano after the 2010 eruption relies on a fourfold methodological approach mixing fieldwork, remote sensing techniques, laboratory analyses and collection of secondary data obtained from local administrations.

(1) The study provides a database documenting lahars occurring from October 26, 2010 to January 25, 2012. The information was

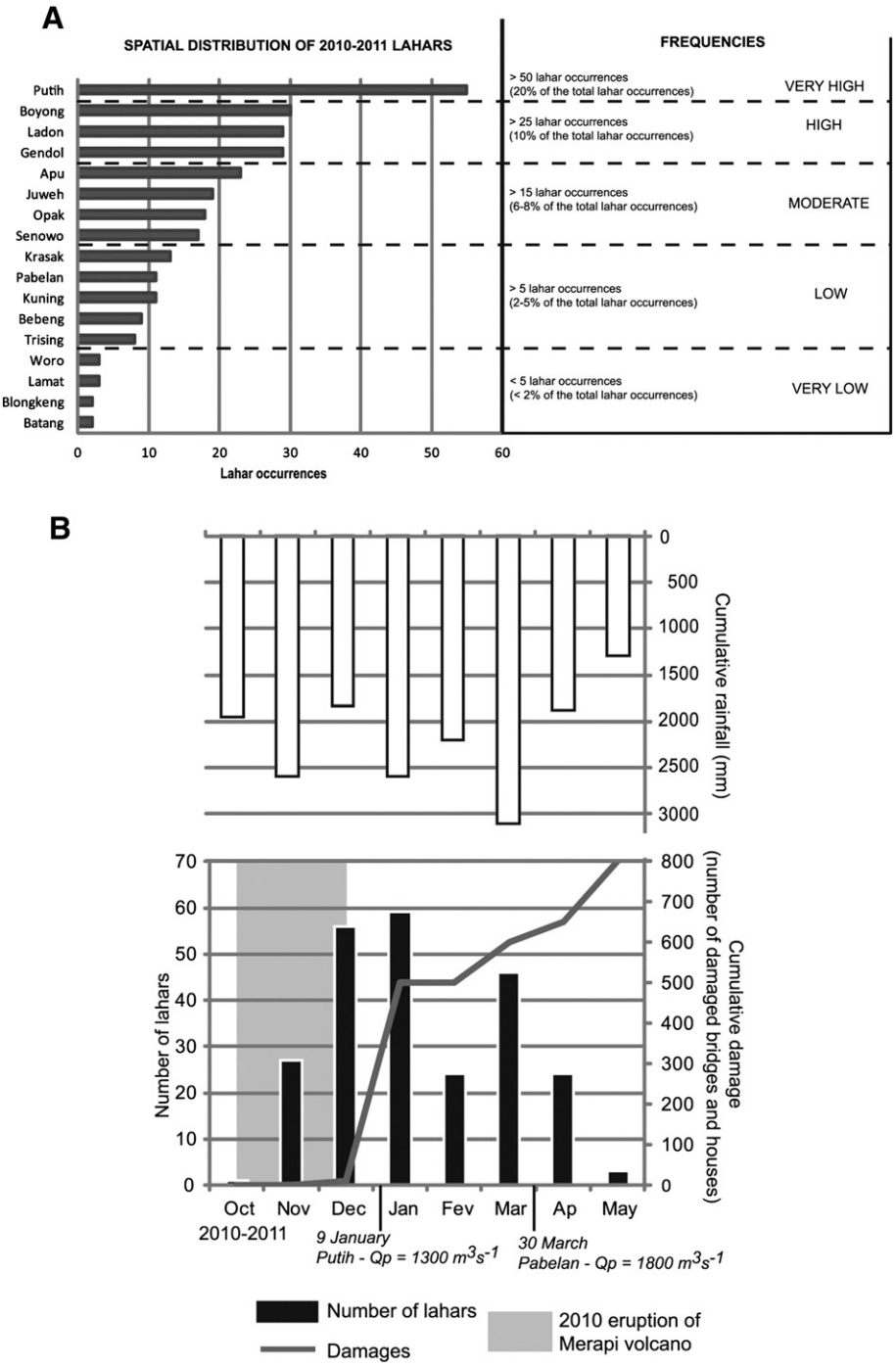


Fig. 2. Frequency and timeline of rain-triggered lahars at Merapi with associated rainfalls and related cumulative damages (October 2010–May 2011).

gathered from various sources, including reports from the Indonesian Office of Volcanology (BPPTK), field observations by researchers and local witnesses, and published accounts from several national and regional Indonesian newspapers. The timeline of lahar events is completed by rainfall data (cumulative rainfalls per month and intensity of rainstorm per hour). Intermittent video records helped to gather some examples of lahars in motion, but continuous recordings of lahars were not possible due to the lack of permanent cameras and operators. The videos were used to estimate discharges of the flows which could be recorded.

(2) Satellite imagery taken before, during and after the eruption (Spot 5 May 17, 2008 and November 15, 2010; GeoEye June 11, 2011) provided data on the planimetric area of lahar sediments. This allowed (1) the mapping of the impacted zones and the main structural damages, and (2) the calculation of deposit areas and volumes in June 2011 (using average deposits thicknesses estimations for each affected river).

(3) Sedimentological data is drawn from field analyses of stratigraphic units in each river basin. Deposit matrix was sampled in 15 locations, and laboratory work provided grain size analyses of the samples, which were described and commented with the sedimentological parameters from Inman (1952) and Folk and Ward (1957). The gravel fraction and the boulders were studied in situ but not sampled.

(4) Secondary data retrieved from the affected municipalities (*desa*) and enquiries with local stakeholders and residents provided valuable information about lahar damages (houses partially or totally destroyed, impacts on bridges, roads and dams, amount of affected people) and about how local communities act to protect themselves against lahars.

3. Results

3.1. A daily hazard after the 2010 eruption on the distal slope of the volcano

Over 240 rain-triggered lahars were recorded during the 2010–2011 rainy season (from October 2010 to May 2011), and 42 at the beginning of the 2011–2012 rainy season (from October 2011 to January 2012). The first lahars occurred on October 27, 2010. They were triggered in the Boyong and Kuning Rivers and remobilized the first pyroclastic-surge deposits (Fig. 1). At least 45 rain-triggered lahars were reported by December 3, 2010. About 70% of the post-eruptive lahars occurred in the Progo River watershed on the west flank of Merapi and the remaining 30% happened in the Opak River watershed on the south flank. Two factors may explain why most of the first post-eruptive lahars were preferentially triggered in the western rivers of the volcano. (1) A higher amount of rain fall from January to April on the western flank (4124 mm at Babadan and Ngepos rain gauges) than on the southern flank (2000 mm at Kaliurang station). (2) 20×10^6 m^3 of fallout tephra which was mainly deposited on the west slopes of Merapi, due to the dominant wind direction during the eruption (Surono et al., 2012).

Frequencies of the 2010–2011 lahars were high (Fig. 2A and B). The Putih River was the most frequently affected by lahars with 55 events reported from October 2010 to October 2011 at a recurrence of approximately two lahars per week during the rainy season (October to March). Due to the broad areal distribution of the 2010 pyroclastic deposits under the active cone, rain-triggered lahars occurred in every basin from the Northwest to the Southeast. For example, on January 9 and on March 4 2011, lahars were reported in 11 rivers around Merapi volcano. On March 30, individual lahars from the Apu, Trising and Senowo Rivers converged in the Pabelan channel generating a much larger lahar (Fig. 2B). The broad distribution of lahars multiplies the risk of disaster, especially when lahars occur simultaneously from different tributaries. Moreover, the runout distance of the 2010–2011 lahars often exceeded 20 km from the summit: lahars generated in the Boyong River reached Yogyakarta City located 24 km south from the Merapi summit on November 29, 2010, March 19 and May 1, 2011. These events have demonstrated that this city is now threatened by lahars as previously suggested by Lavigne (1999).

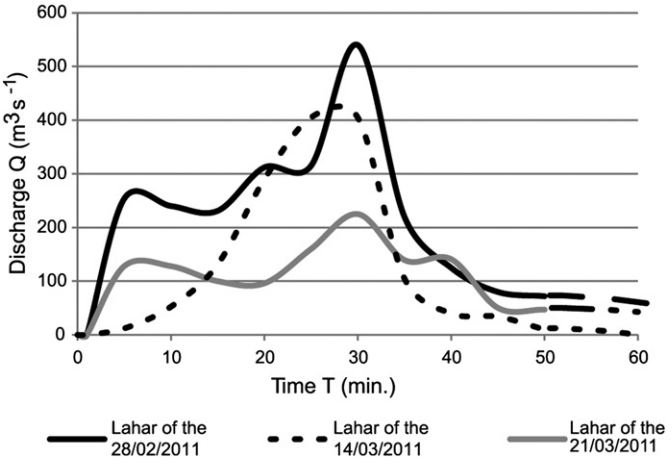


Fig. 3. Discharges of three lahars on the distal Gendol river (February–March 2011), from video data. Discharges were estimated from the initiation to the attenuation of the flows.

Recordings of lahars in motion on the distal slope of the volcano show irregular and multipeaked discharges. Due to limitations in equipment availability it has not been possible to continuously record all lahars. However, we could study some events in the Gendol River. The behavior of each single flow itself is not regular: each peak of the hydrographs for three of these lahars corresponds to a flow pulse (Fig. 3). For example, on February 28, 2011, after the front discharge reaches $Q=250$ m^3 s^{-1} (from video recording), it undergoes a slight decrease in intensity after only 10 min (Fig. 3). The same pattern of peak pulse followed by decrease of discharge occurs again between T (time when the lahar reaches the recording point) + 20 to T + 30, corresponding to the peak flow reaching $Q_p=540$ m^3 s^{-1}. Similar pulses can be observed during the March 21 event with lower discharges (peak flow rate $Q_p=225$ m^3 s^{-1}). Although these two flows were distinguished by rapid increase in river discharge, the March 14 hydrometric record illustrates that lahars on the distal slope of Merapi may also exhibit more gradual increases in flow (Fig. 3). In all three cases the peak flow did not coincide with the frontal passage when there was one, and the maxima of the recorded flows were all recorded at T + 30.

3.2. Geomorphic impacts of lahars and related damages on the distal slope

3.2.1. Lahar corridors

During the 2010–2011 rainy season, river channels were affected by lahars and by their morphogenic processes: riverbank erosion, channel widening and riverbed downcutting. Narrow rivers of the distal slope of the volcano which had not been affected by lahars for almost forty years rapidly transformed to large corridors, which generation represents a risk in this densely populated area (950 inhabitants/km^2). Examples of the creation of lahar corridors will be shown for the Opak River (south slope) and for the Pabelan River (west slope).

The distal part of the Opak River was transformed very rapidly by repeated lahars (17 occurrences) from a narrow stream to a wide corridor, which threatens villages and crops located on the riverbanks (Fig. 4). Over the 20th century, few lahars occurred in the Opak River; the river is not even represented as a lahar-prone area on the 2006 and 2011 hazard maps. Before the 2010 eruption, the distal part of the Opak River at Panggung and Teplok villages was a small stream between 1.5 m and 2 m deep and 2 m wide. The tributary which flowed along the western part of the Panggung and Teplok villages was captured by the Opak in December 2010, shortly after the beginning of lahar occurrences (Fig. 4). This reorganization of the drainage pattern at a local scale is due to the rapid generation of a large corridor. Only 9 occurrences

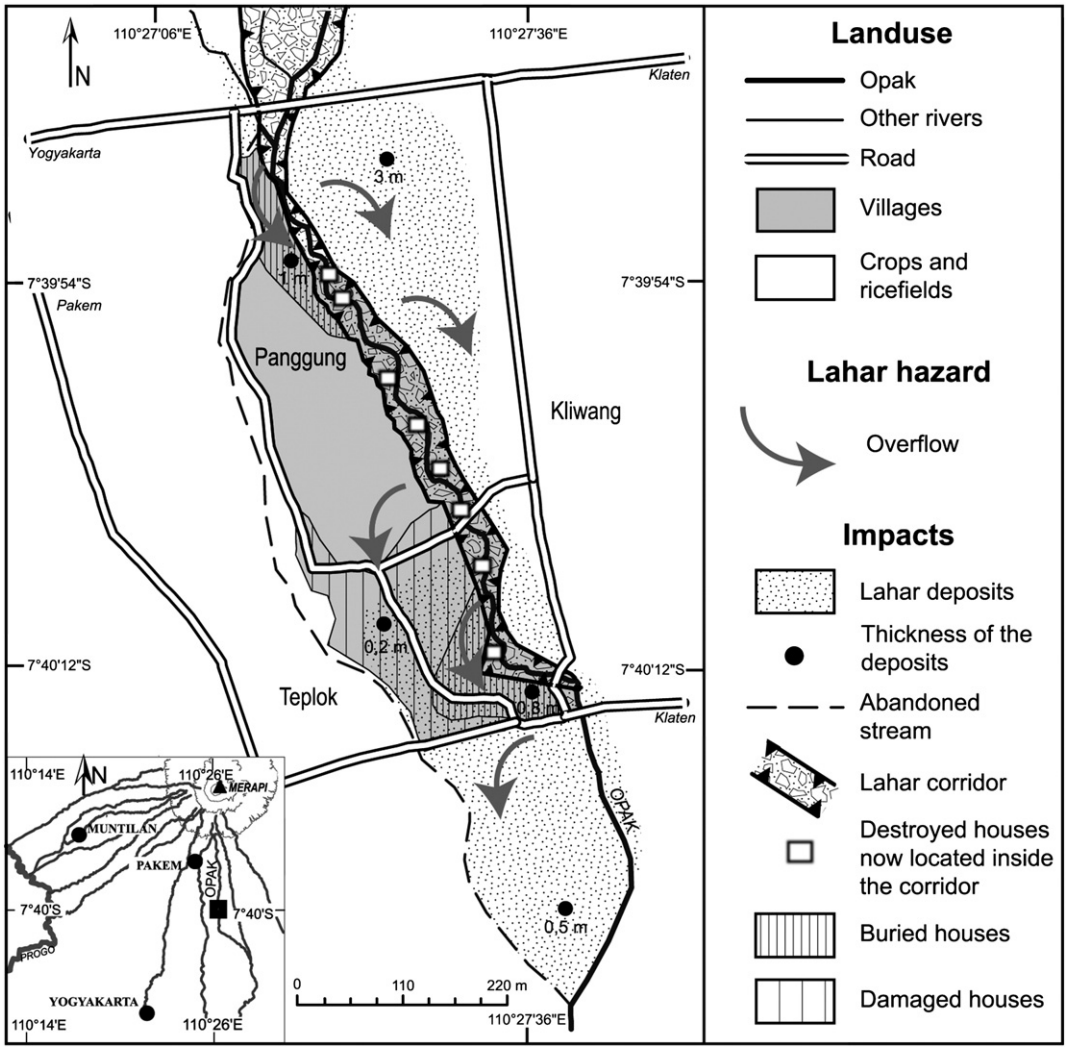


Fig. 4. Formation of the Opak lahar corridor on the distal slope of Merapi.

occurring in 3 months (November 2010 to January 2011) created a 4 m deep and 20 m wide corridor. It expanded eastward into ricefields, and westwards into the villages: 8 houses built near the river were gradually located inside the lahar corridor as it formed, and were finally destroyed (Fig. 4). Moreover, 18 houses and buildings inside the villages were exposed to overflows from lahars until May 2011. For example on March 22, 2011, a lahar with peak discharge $Q_p = 245\ m^3\ s^{-1}$ (estimated from the average surface velocity, width and depth of the flow) flew over the villages. The depth of the mud reached 0.1 m to 1 m and damaged 2 ha: crops were buried, houses were partially destroyed, and furniture and electronic goods were lost.

Contrary to the Opak River, the distal part of the Pabelan basin is constrained at the bottom of a 20 m-deep valley which is a former corridor reactivated by lahars which occurred in 2010–2011 (Fig. 5). A part of the Sidoharjo village was built on a 2.5 m-high volcaniclastic terrace overlooking the riverbed, and was totally destroyed on March 30, 2011 by the most voluminous event recorded during the 2010–2011 rainy season. During the first six months following the 2010 eruption, seven lahars were reported in this river, but they were restricted to the channel. Late afternoon on March 30, heavy rains (269 mm total and an estimated intensity between 40 and 52 mm/h) were reported on the north and west flanks of Merapi in the late afternoon. Three lahars occurred simultaneously in the tributaries of the Pabelan River: Apu, Trising and Senowo Rivers. At 17:00, Babadan observatory (4 km from the crater) issued a warning that a large lahar flowing along the Senowo River would join a second lahar flowing along the Trising. At 17:55, Sidoharjo hamlet, located 12 km downstream from the observatory was devastated by the first lahar wave which had overtopped the riverbed. This was followed by another wave 45 min later. The 2.5 m-high terrace was covered by 4.5 m of deposits, showing that the lahar reached a maximum depth of 7 m (Fig. 6). The peak discharge of this event was estimated at 1800 $m^3\ s^{-1}$, making it one of the largest lahar ever recorded at Merapi volcano (2000 $m^3\ s^{-1}$ in the Putih river in 1985, Jitousono et al., 1995). There were no fatalities but 19 houses of the village were entirely destroyed, and one of the bridge piers of a bridge was transported 950 m downstream (Fig. 5). The Pabelan lahar suggests that large lahars are a cause for concern in distal areas of heavily branched watersheds as they can reach high depths (>5 m) and discharges, generating massive overflows. Parts of villages located on ancient terrace inside the corridors can thus be destroyed by lahars, as exemplified by the Sidoharjo event (Fig. 5).

3.2.2. Lahar-related avulsions

Lahars can generate avulsions (sudden shift of the river channel) on the distal slope of Merapi volcano, potentially creating major disasters on densely populated areas. The example of the Sirahan village located on the Putih River (west flank) will be detailed. On the Putih River 19 lahars damaged the Sirahan village in November and December 2010. This settlement is located on the west volcaniclastic ringplain of the volcano where the valleys are not deeply cut. Lahars in November and December 2010 brought more than 5 m of volcaniclastic deposits in the riverbed. Meanwhile, riverbank erosion

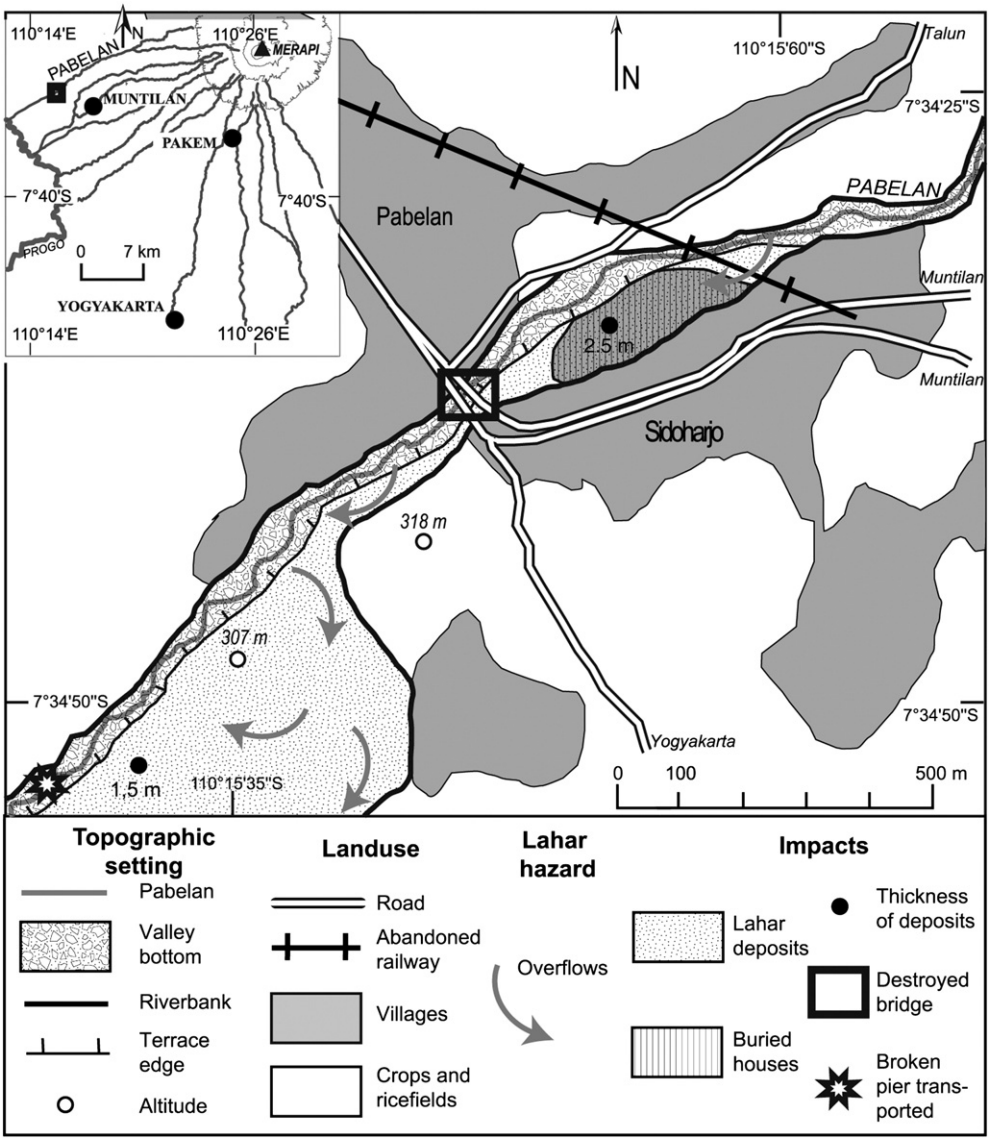


Fig. 5. The Pabelan lahar corridor on the distal slope of Merapi.

reached 10 m, and created a corridor. To avoid overflows, levees were built using sandy volcaniclastic material deposited by lahars in order to protect the villages. However, those levees have proven to be largely ineffective in case of voluminous lahar, as exemplified by the January 9, 2011 lahar. Rain-gauges from the observatory posts of the volcano located on the west flank of the volcano (Babadan and Ngepos) recorded a total of 140 mm rainfall from 16:50 to 20:05, with an average rainfall intensity of 28 mm/h in the headwaters of the Putih River. At 18:00, a lahar was reported at Ngepos observatory (Fig. 2), and as it passed under the bridge deck recordings from the Office of Volcanology (BPPTK) it had a recorded depth of 5 to 6 m and a discharge of $Q_p =$ 1300 $m^3\ s^{-1}$ (estimated from the average surface velocity, width and depth of the flow). At 18:40 pm, the lahar reached Sirahan and destroyed the levees. It accumulated in the rice fields, while another part followed the natural slope and flowed along the road, which was totally destroyed as the lahar cut a new 3.5 m-deep channel in the middle of the inhabited area, before returning to the Putih river 800 m downstream (Fig. 6). A third channel temporarily connected the Putih to the Blongkeng River, and had turned the north road of the village into a stream for 4 months. The same avulsion process occurred again for all of the 25 following lahars from January to May 2011, causing severe damages in Sirahan as lahars flowed in the new channel created inside the village (Fig. 6). As a result, 254 houses were damaged, 37 of them were completely destroyed, and 30 ha of crops were buried under 3 m of deposits (Fig. 7). Main damaged are localized near the avulsion channels. At the end of the 2010–2011 rainy season, the Putih River flowed along a 30 m-wide corridor.

A similar event occurred on May 1, 2011 affecting the Ngerdi village along the Gendol River (Fig. 5). This lahar, generated by high rainfall intensity (41 mm/h) reached Ngerdi at 6.10 pm with approximately 2 m deep. It did not follow the sinuosity of the former riverbed but rather pushed straight through the ricefield and the village (Fig. 6). About 40,000 m^2 of crops were buried and 51 houses were severely damaged. As a consequence the Gendol River was divided into two separate channels, including the new 15 m-wide and 1.5 m-deep channel within the Ngerdi village.

3.2.3. Typology of related damages

During the 2010–2011 rainy season lahars damaged 860 houses on the distal slope of Merapi including Yogyakarta City (Fig. 8), destroyed 14 sabo-dams and 21 bridges, cut the main road from Yogyakarta to Semarang and buried at least 70 ha of land. Table 1 shows the damages related to the main geomorphic process related to lahars (riverbank widening, riverbed downcutting, avulsions and overflows). Damaged houses were mainly located on the west flank of Merapi volcano, along the Putih River in Jumoyo and Sirahan villages (Fig. 8). In Jumoyo, all the houses located near the Yogyakarta–Semarang highway were impacted by lahar

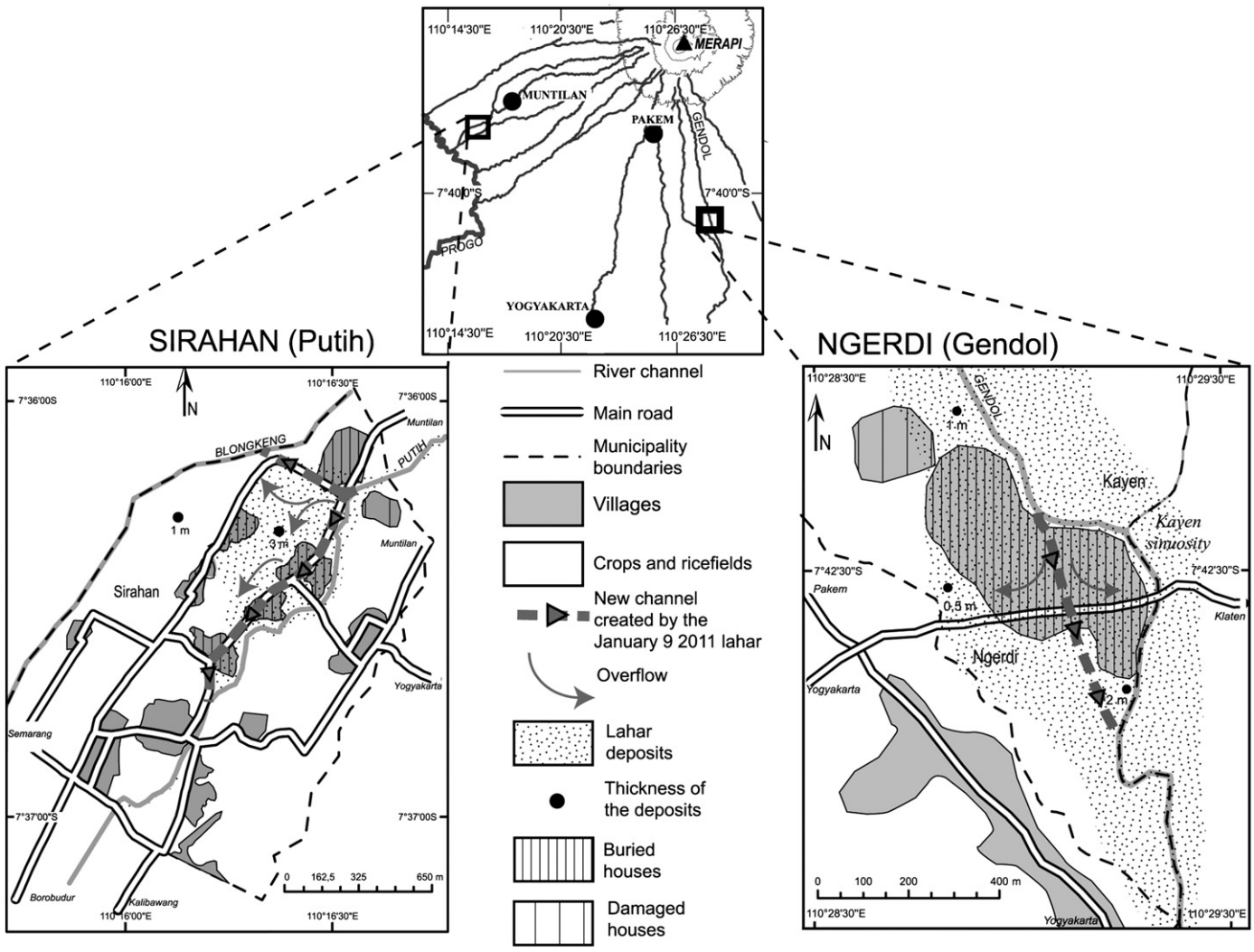


Fig. 6. Avulsions on the distal Putih and Gendol Rivers.

overflows: 20% of them were totally destroyed and 80% of the houses lost at least one wall. The highway passing Jumoyo road was crossed 15 times by lahars since January 2011. Whilst the road was being cleared, traffic had to be diverted to the Menoreh Mountains, located on the west side of the Progo River, generating traffic-jams on a narrow and sinuous mountain road. More than 3000 people were affected by lahars, and shelters initially opened during the eruption were reopened in order to accommodate lahar victims.

Table 1
Lahar morphogenic processes and related damage.

Process	Damage	Examples
Avulsion	Villages almost entirely damaged (>100 houses)	Sirahan (Putih)
		Jumoyo (Putih)
	Road destroyed on long distances (>200 m)	Sindumartani (Gendol)
	Crops and ricefields buried (>4 ha)	
Riverbank erosion and corridor widening	Villages partially damaged (>20 houses)	Panggung (Opak)
		Pabelan
	Bridges destroyed	
Overflow	Villages punctually damaged (<20 houses)	Pabelan
		Putih
	Road destroyed on short distances (<100 m)	Boyong/Code
		Opak
	Crops and ricefields buried (<4 ha)	Gendol

3.3. Lahar deposits

3.3.1. Main lithofacies of 2010–2011 volcaniclastic deposits

Lahar corridors are composed by volcaniclastic materials brought by lahars; analyses of the main lithofacies from the proximal to the distal slopes can help to understand better the extension of hazard-prone areas. Proximal facies (<6 km from the summit) contain layers with intermediate-size boulders (15 cm to 30 cm in units S3 on the Senowo river, T2 on the Boyong river — Fig. 9A). Units are poorly sorted and they typically do not present any internal organization. A rough reverse grading can be seen on the layer T4 and bedded subsets are identifiable in layer T4 (Fig. 9A). Here, proximal deposits lay on top of 2010 pyroclastic deposits (T1), with metric boulders (>1 m) supported by a fine sand matrix. In all cases, the proximal deposits were located within steep-sided valleys (>20 m) and, as a consequence, risk of overflow remains low.

Medial deposits (6 to 15 km from the crater) contain units characterized by coarse-grained gravels in a sand–pebbly matrix (K2 on the Boyong river, B1 on the Gendol river — Fig. 7A) or centimetric to decimetric boulders (K1, B2). Units were poorly sorted and did not present any grading or bedding. The “Kemiri” deposit (Fig. 9A) contains a thin bed of clast-supported gravels at the base (K1) which was covered by two layers: clast-rich K2 and the more dilute K3. In contrast, the “Bronggang” deposit was associated with a more powerful event. The first pulse deposited

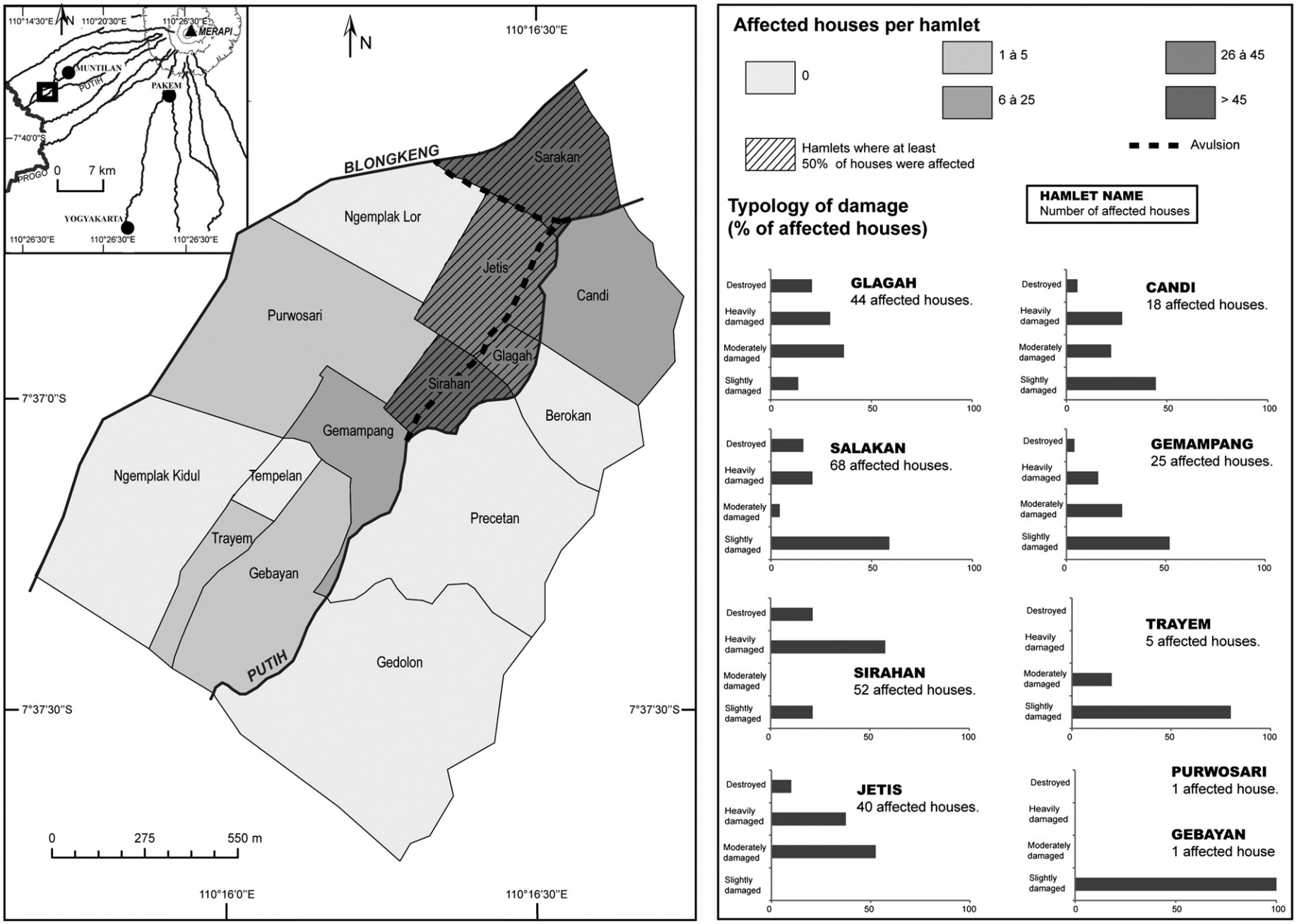


Fig. 7. Impacts of the avulsion on houses in Sirahan village, on the distal Putih River.

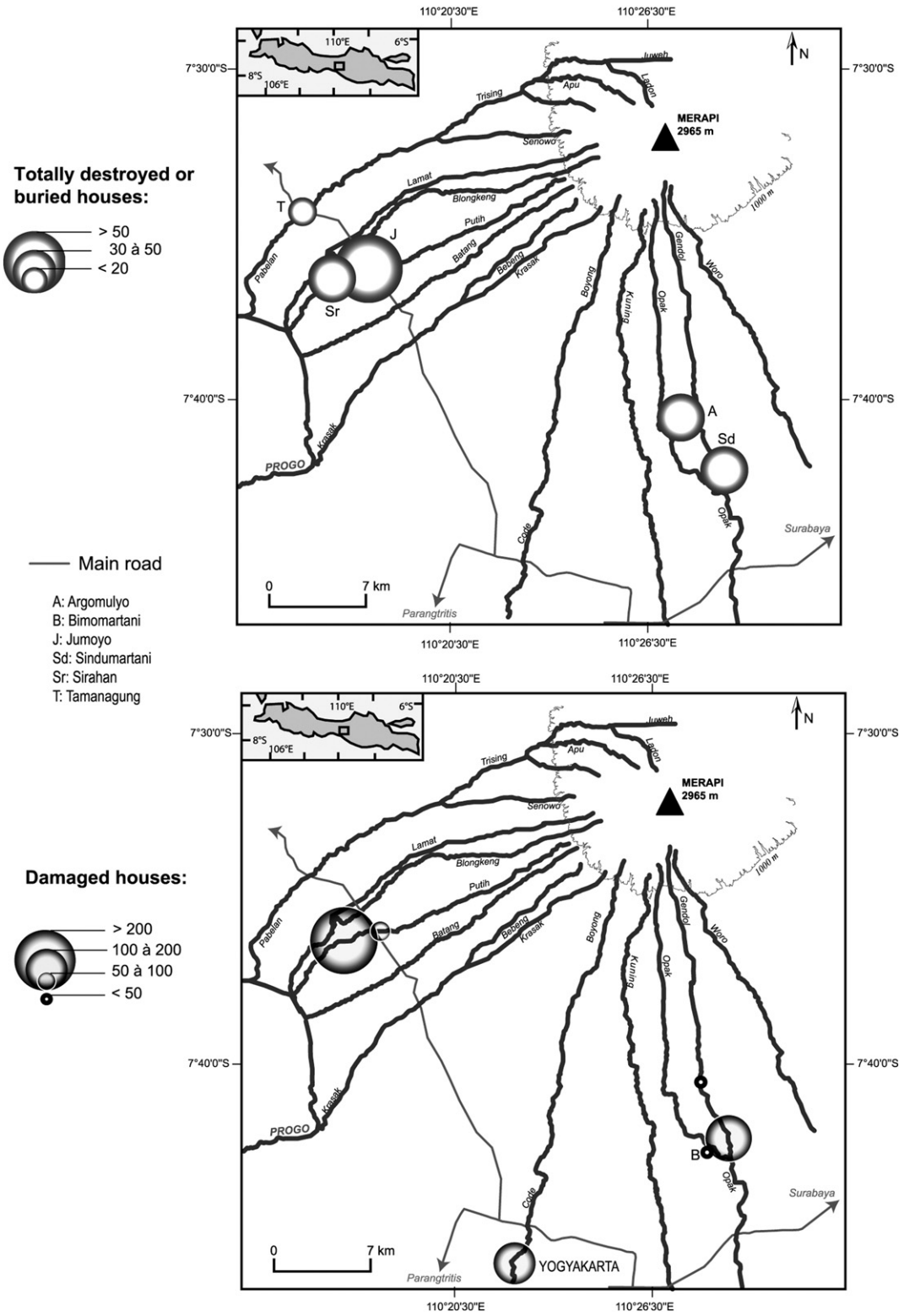


Fig. 8. Lahar damage on houses around Merapi. Note that the affected houses are located on the distal slope of the volcano.

pebbles and cobbles in a coarse-grained sand matrix, and the second pulse transported and then deposited boulders in a gravel matrix. The deposit thickness and the presence of coarse materials in the "Bronggang" section suggest that lahars have the potential to be very damaging, especially as the flow can easily spill over on the adjacent areas as there are no steep riverbanks as in the proximal area. Far-reaching pyroclastic deposits (>15 km to the crater) on the Gendol river increase the risk as there is no lack of available materials to be remobilized by lahars. In the medial Boyong River, deposits (as in the "Kemiri" section) are commonly matrix-supported (Fig. 9A) with very few large clasts <−6 φ (64 mm).

On the ringplain surrounding the volcano, distal deposits are more diverse than those observed in the proximal and medial locations. The "Pabelan" deposit was emplaced by the most voluminous lahar which occurred at Sidoharjo (see Section 3.2.1). Debris-flow structures can be recognized in the layers Pa2 and Pa3, with large boulders taken supported by a coarse-sand or gravel matrix (Fig. 9A). We infer from field observations that the fine-grained unit at the top (Pa4) was deposited during the peak discharge after the first pulse corresponding to the front of the flow. The "Opak" section has fine sand (Op1) and silt-sized materials (Op2) likely deposited by more dilute lahars. Such fine material might have come from fine 2010 pyroclastic deposits located in the headwaters of the Gendol River. Finally, the "Putih" deposit contains cross-bedded layers of gravel, coarse sand, and fine sand (Fig. 9A), and suggests water-rich flows reaching this location. Distal deposits commonly contain finer material than deposits in proximal and medial locations (Fig. 9B and C). From these observations, we can infer that adjacent areas of the distal Pabelan corridor (>20 km from the summit) are exposed to voluminous lahars carrying large boulders. However, deposits on the Opak and Putih distal corridors show evidences of more dilute events with lower destructive potential.

3.3.2. Grain-size analysis

Grain-size analysis was realized in order to assess better the sedimentary composition of the lahar corridors at Merapi; it shows different types of materials: proximal–medial vs. distal deposits, and south-west deposits vs. south deposits.

Matrix samples ($n=15$) were taken all around the volcano (grain diameter > -1 φ, e.g. <2 mm ($\varphi = -\log_2 d$, with d: grain size in mm), but gravels and boulders were not sampled). Sedimentological analysis (Fig. 10A; Table 2) revealed materials to be moderately well-sorted to very poorly sorted (standard deviation between $\sigma_\varphi = 0.68$ and $\sigma_\varphi = 2.23$), with only three samples showing moderate sorting. Most of the sample were coarse to strongly coarse-skewed ($Sk < -0.5$ φ) showing excess coarse material (Fig. 10C and D). Only one sample, taken on the Krasak River, is characterized by fine sand grain size (mean grain size $Mz = 2.1$ φ). Other samples contain coarse sand (0 φ$< Mz <$ 1 φ) and all show a small proportion of clay (>8 φ, e.g. <0.004 mm, with average cumulative percent weight reaching only 1.6%). Overall, there is high homogeneity between lahar deposits from all locations around the volcano, and deposits are typical of non-cohesive debris flows and hyperconcentrated flows with poorly-sorted material and coarse-grained sand matrix with clay content <3% (Scott et al., 1995; Capra et al., 2004). Moreover, most of the matrix samples show a high proportion of coarse sand. Kurtosis (K_G) analysis illustrates the statistical prevalence of coarse sand (Fig. 10D), as ten samples show a leptokurtic distribution ($K_G > 1.11$); this characteristic has already been found at Merapi by Lavigne and Thouret (2002), in hyperconcentrated flows facies.

The main difference which can be distinguished between sampled deposits is in the matrix composition. The skewness (*Sk*) and the mean size (*Mz*) of the samples tend to decrease with increased distance from the summit (Figs. 9C and 10A and C). A strongly coarse-skewed sediment distribution was found only in proximal and medial deposits. In contrast, normal and fine-skewed distributions characterizes only in

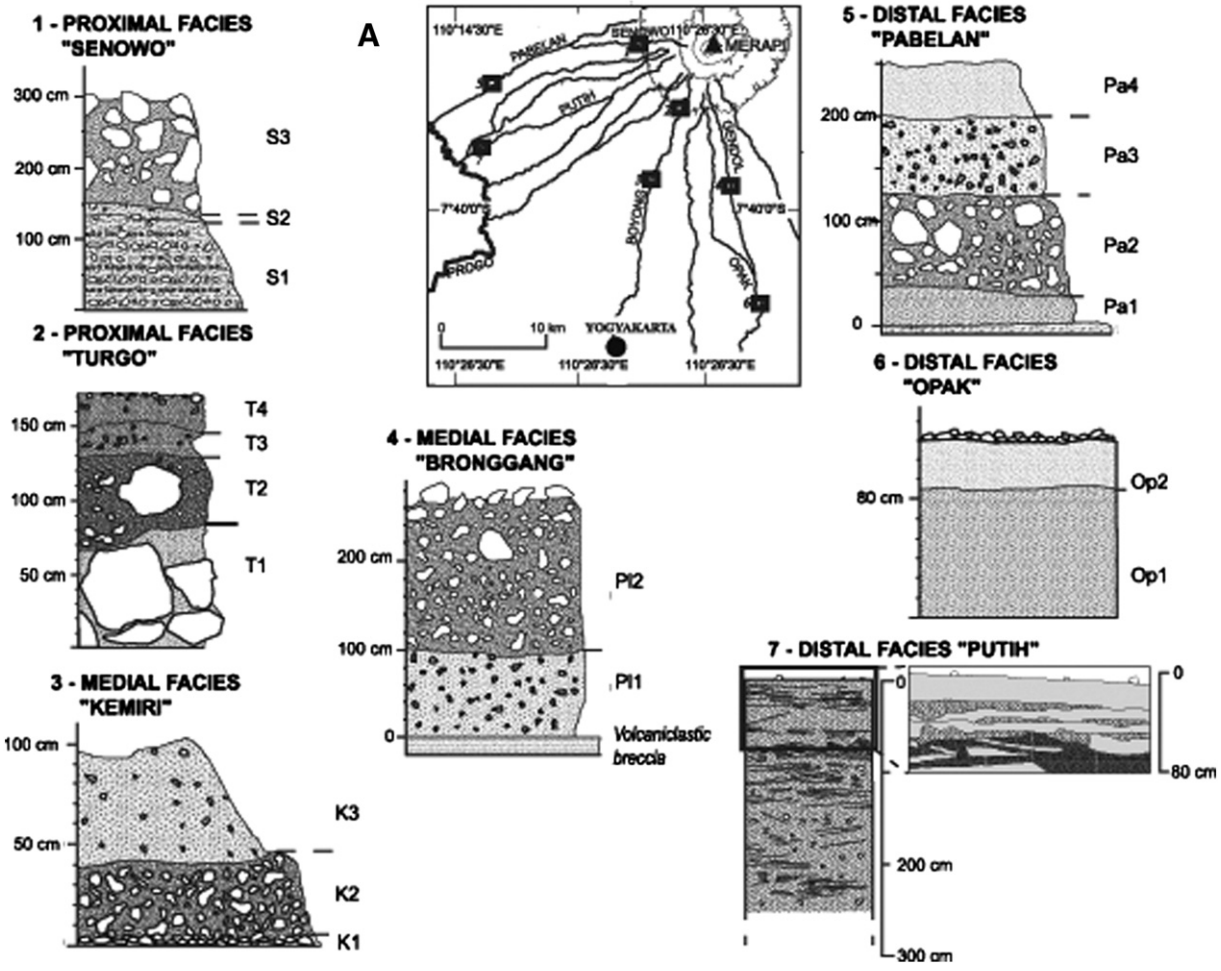


Fig. 9. A and B: Sedimentary facies of 2010–2011 rain-triggered lahar deposits at Merapi. C: Comparative cumulative grain-size curves of lahar deposit facies.

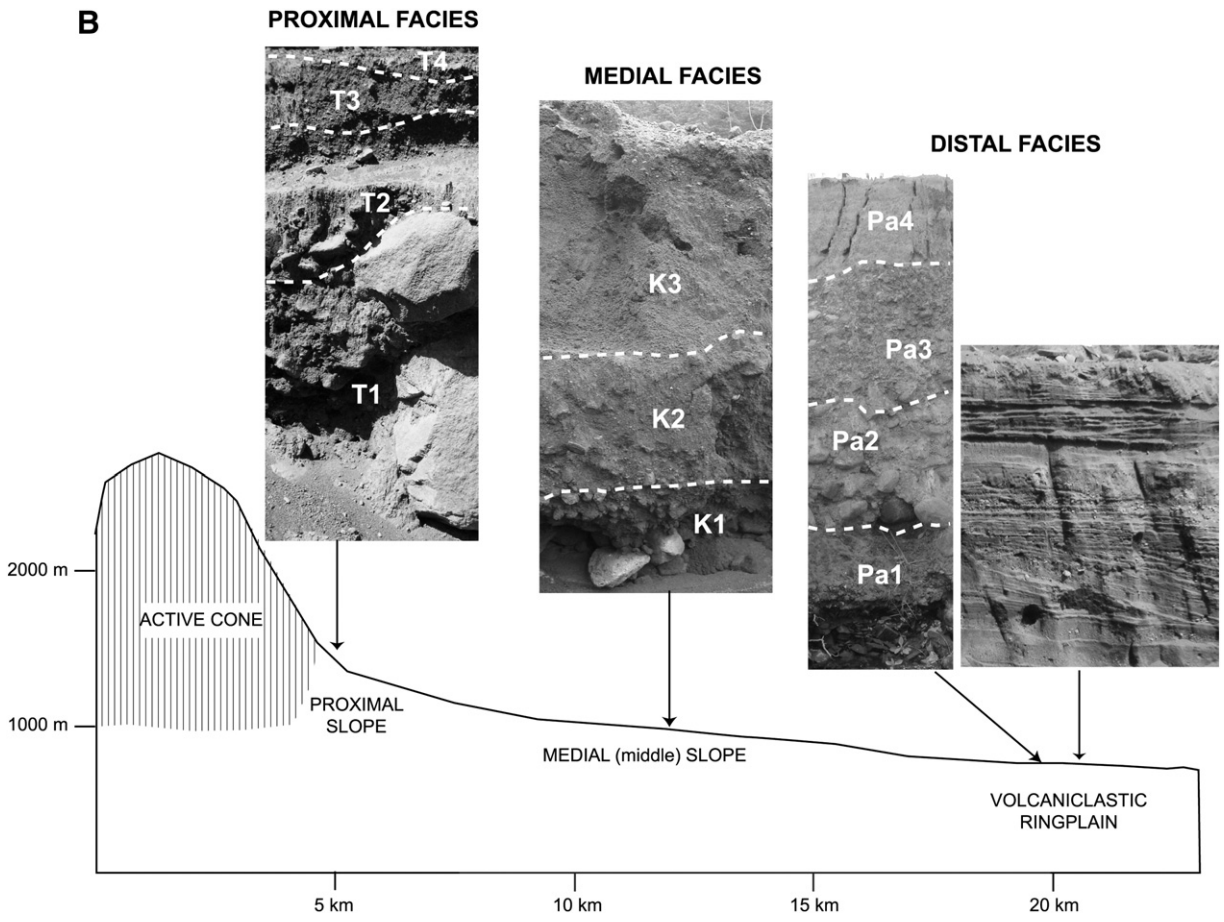


Fig. 9 (*continued*).

deposits along the distal slope. This difference is more evident when considering mean grain size. All deposit samples from the proximal and medial slopes were typically composed of coarse sand, while deposit samples from the distal slope contained fine-grained sand matrices. This likely reflects the dilution of flow downstream after it has already deposited the bulk of the coarser sediment. This has been observed before on Merapi and on other composite volcanoes (Pierson and Scott, 1985; Pierson, 1995; Cronin et al., 1997; Lavigne et al., 2000a,b; Lavigne and Thouret, 2002).

Table 2
Sedimentological characteristics of matrix samples.

	Location	River	Distance from summit	*Mz* (φ)	σ (φ)	K_G (φ)	Sk (φ)
Proximal	Kemiren	Bebeng	5	0.54	1.44	1.01	−0.18
	Balerante	Woro	4	0.68	2.23	1.21	−0.41
Medial	Sudimoro	Bebeng	9	0.43	1.73	1.27	−0.35
	Kemiricilik	Boyong	6	0.08	0.90	0.96	−0.19
	Kemiri	Boyong	8	0.47	2.19	1.38	−0.48
	Manggong	Gendol	10	0.62	2.13	1.14	−0.43
	Srumbung	Putih	12	0.54	1.77	1.28	−0.33
	Cangkringan	Kuning	13	0.62	1.72	1.22	−0.31
	Bronggang	Gendol	15	0.68	1.83	1.29	−0.27
Distal	Opak–Gendol	Opak	21	0.42	1.05	0.98	−0.11
	Jambon	Gendol	18	1.03	2.02	0.95	−0.22
	Prambanan	Opak	24	1.21	1.05	0.93	0.12
	Pondokrejo	Krasak	18	2.10	1.71	1.35	0.16
	Jumoyo	Putih	16	1.14	1.93	1.21	−0.15
	Sukorini	Woro	25	1.60	1.12	1.12	0.04

Mz: mean grain size = (Q16 + Q50 + Q84)/3.
σ: sorting (standard deviation) = (Q84 − Q16)/4 + (Q5 − Q95)/6.6.
K_G: peakedness (graphic kurtosis) = (Q95 − Q5)/2.44(Q75 − Q25).
Sk: skewness = (Q16 + Q84 − 2Q50)/(Q84 − Q16).

Another distinction between deposits of the south-western flank and other deposits is the difference in cumulative grain-size frequencies of the matrix (Fig. 11A). Southwestern matrix deposit on the medial slope of the Putih and Krasak rivers are slightly finer-grained than the southern deposits (Gendol and Boyong), with mean size *Mz* reaching 1.1 φ (Putih) and 2.10 φ (Krasak). Fine fallout tephras covering the western flank of the volcano may constitute a high amount of the reworked material observed in the 2010–2011 lahar deposits, but the difference between deposits on western and southern slopes is limited to watersheds draining the southwestern part of the volcano (Putih and Krasak Rivers). Grain size histograms (Fig. 11B) of the Putih lahar deposits from present a peak in fine sand (4 φ), and the deposits contain no large boulders (< −6 φ). In contrast, histograms from medial slopes of the Gendol and Senowo lahar deposits (Fig. 11B) have higher proportions of coarse materials (boulders to gravel) with a peak in boulders (< −6 φ), and very few fine sand (2 to 4 φ) and silts (>4 φ).

In summary, analysis of lahar deposits provides key information for understanding lahar-related hazards. Even though distal areas of the volcano seem to be frequently threatened by more dilute lahars, discrete high-magnitude events also threaten areas located more than 15 km from the crater. Lahar deposits are typically clast-rich in most rivers, except on the southwestern part of the volcano. Results from the Senowo and Gendol sample deposit histograms show that high-magnitude and potentially destructive lahars carrying very coarse materials can be triggered on those rivers for the next decade after the 2010 eruption at Merapi.

3.3.3. Lahar deposits: hazard and resource

The risks associated with lahars at Merapi are not spatially restricted to villages near lahar-prone rivers, but also to valley bottoms covered with volcaniclastic deposits, which represents an

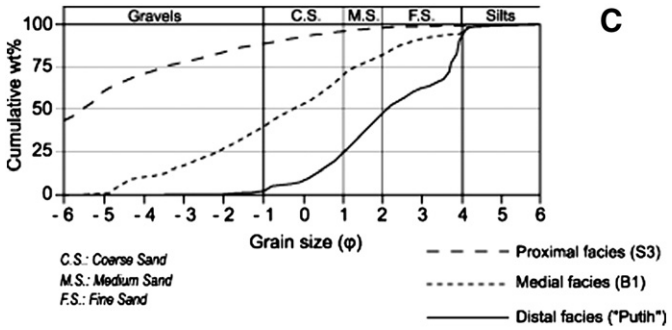


Fig. 9 (*continued*).

economically valuable building material resource. Using the calculation of deposit areas and volumes in June 2011 from satellite imagery and the average deposits thicknesses estimations from the field for each affected river, the total volume of lahar deposits can be estimated at 0.025 km^3 (Table 3). The Putih and the Gendol lahar corridors are the main deposit areas with respective volumes estimated at 4.9×10^6 m^3 and 3.6×10^6 m^3, and contain the most exploited quarries at Merapi with 1674 workers (63% the total number of workers estimated at more than 2600 people per day, Fig. 12). Quarrying sand and boulders from volcaniclastic sediment in lahar corridors is, however, quite dangerous. Villagers are dependent on a resource which is brought by the hazard. For twenty years, hundreds of trucks and thousands of workers have traveled daily through areas of high lahar hazard. The difficult socio-economic conditions around Merapi volcano limits people's livelihood to working in the quarries, which remain poorly controlled by the government (De Bélizal et al., 2011). More than 2000 people quarry deposits every day (De Bélizal, 2012), because they can earn four times the daily income of a farmer. Risks are particularly high on the quarries located in the Putih lahar corridors where hazard is very frequent, and in the Gendol lahar corridor where more than 1000 people quarry volcaniclastic deposits every day (De Bélizal, 2012). Following the 2010 eruption, the exposure of workers to lahars has increased greatly: among the 3 people killed by the 2010–2011 lahars, 2 of them where sand miners. Thus, when managing for lahar hazard, it is important to remember that lahars at Merapi volcano bring a valuable resource to communities, which are ready to increase their exposure to hazard by quarrying deposits on lahar corridors. Risk managers should not only focus on villages threatened by lahars: they should also take into consideration the high number of people working every day in lahar-prone areas.

Table 3
Estimation of lahar deposits volumes.

N°	Section	Deposition area (m^2)	Average deposit thickness (m)	Volume (m^3)
1	Ladon	30,280	0.8	24,224
2	Juweh 1	65,440	0.8	52,352
3	Juweh 2	201,749	1	201,749
4	Apu	450,370	3.1	1,396,147
5	Trising	561,713	1.4	786,398.2
6	Senowo 1	262,220	1.1	288,442
7	Senowo 2	528,698	2.2	1,163,135.6
8	Pabelan 1	997,446	2.7	2,693,104.2
9	Pabelan 2	750,705	0.4	300,282
10	Lamat 1	39,452	0.9	35,506.8
11	Lamat 2	62,822	1.4	87,950.8
12	Lamat 3	75,633	1.7	128,576.1
13	Lamat 4	14,606	0.5	7303
14	Lamat 5	14,656	1.1	16,121.6
15	Lamat 6	13,257	0.3	3977.1
16	Blongkeng 1	145,714	0.8	116,571.2
17	Blongkeng 2	102,087	1.2	122,504.4
18	Blongkeng 3	62,923	0.5	31,461.5
19	Blongkeng 4	281,453	0.4	112,581.2
20	Putih 1	357,781	1.4	500,893.4
21	Putih 2	363,591	2.6	945,336.6
22	Putih 3	515,241	1.1	566,765.1
23	Putih 4	687,477	2.1	1,443,701.7
24	Putih 5	413,968	3.5	1,448,888
25	Putih 6	76,000	0.3	22,800
26	Batang	169,641	0.4	67,856.4
27	Bebeng 1	691,742	0.7	484,219.4
28	Bebeng 2	384,668	1.4	538,535.2
29	Bebeng 3	114,025	1.6	182,440
30	Krasak 1	423,178	1.1	465,495.8
31	Krasak 2	651,643	0.7	456,150.1
32	Boyong 1	238,395	1.1	262,234.5
33	Boyong 2	269,820	1.9	512,658
34	Boyong 3	1,407,809	0.6	844,685.4
35	Code	846,508	0.3	253,952.4
36	Kuning 1	259,042	0.8	207,233.6
37	Kuning 2	621,122	1.7	1,055,907.4
38	Kuning 3	810,812	0.6	486,487.2
39	Opak 1	364,070	1	364,070
40	Opak 2	1,187,326	1.3	1,543,523.8
41	Gendol 1	675,549	1.2	810,658.8
42	Gendol 2	1,216,167	2.3	2,797,184.1
43	Woro 1	668,472	0.7	467,930.4
44	Woro 2	473,986	1.2	568,783.2
45	Woro 3	514,707	1	514,707
		Total		25,381,485.2 m^3 (0.025 km^3)

3.4. Managing lahar hazard at Merapi volcano after the 2010 eruption

Despite the frequency and extent of lahars, there have been few human casualties (3 killed and 15 hurt). In every watershed a spontaneous and community-based oversight of the river conditions was developed shortly after the eruption. It involves well-trained volunteers who learned to recognize the conditions that can trigger lahars, and who can very quickly send the order to evacuate threatened people. Some dwellers obtained radio transmitters from NGO's during the 2010 eruption, and built *poskos* (look-out stations). For example, the Opak and the Gendol Rivers have *poskos* near most villages located 10–18 km to the crater, where avulsions and overflows are to be expected (Fig. 13). Upstream look-out stations are only dedicated to the monitoring of the valley, whereas downstream *poskos* also have to issue the alert and the order of evacuate if needed.

This self-organization allows villager to obtain pertinent information about the upper part of the watershed and self-evacuate when a lahar is reported. They recognize early warning signs of a lahar in the headwaters weather conditions. With radio transmitters, communities can receive reports from seismometers located around the volcano. When heavy rainfall occurs, the ground vibration produces a recognizable signal alerting watchers of possible lahar activity, and allowing them to issue evacuation orders. This type of self-organization is found on almost all river communities, mainly on the Putih, Boyong, Opak and Gendol Rivers, where many look-out stations have been built in upstream and downstream locations. Thus, people living in hazard-prone areas can be warned at least 15 to 30 min before the arrival of a lahar (Fig. 13).

Three main factors are responsible for the success of this self-management: (1) many young people living in villages threatened by the eruption took part in an evacuation process organized and developed by the government (Mei and Lavigne, in press). So, many of them worked as member of SAR (Search And Rescue) teams, and have helped authorities evacuate people and search for wounded or the dead. They became accustomed to the instrumental monitoring of the volcano, and in particular the alarm call issued by seismometers from BPPTK. This great solidarity between BPPTK and SAR teams began during the 2010 eruption: rescuers on the slopes of the volcano used to listen to the information issued by BPPTK (P. Jousset, written communication). As the eruption came to an end, some of these volunteers monitored the rivers for lahars. (2) An alert can be spread quickly in a community because almost everyone

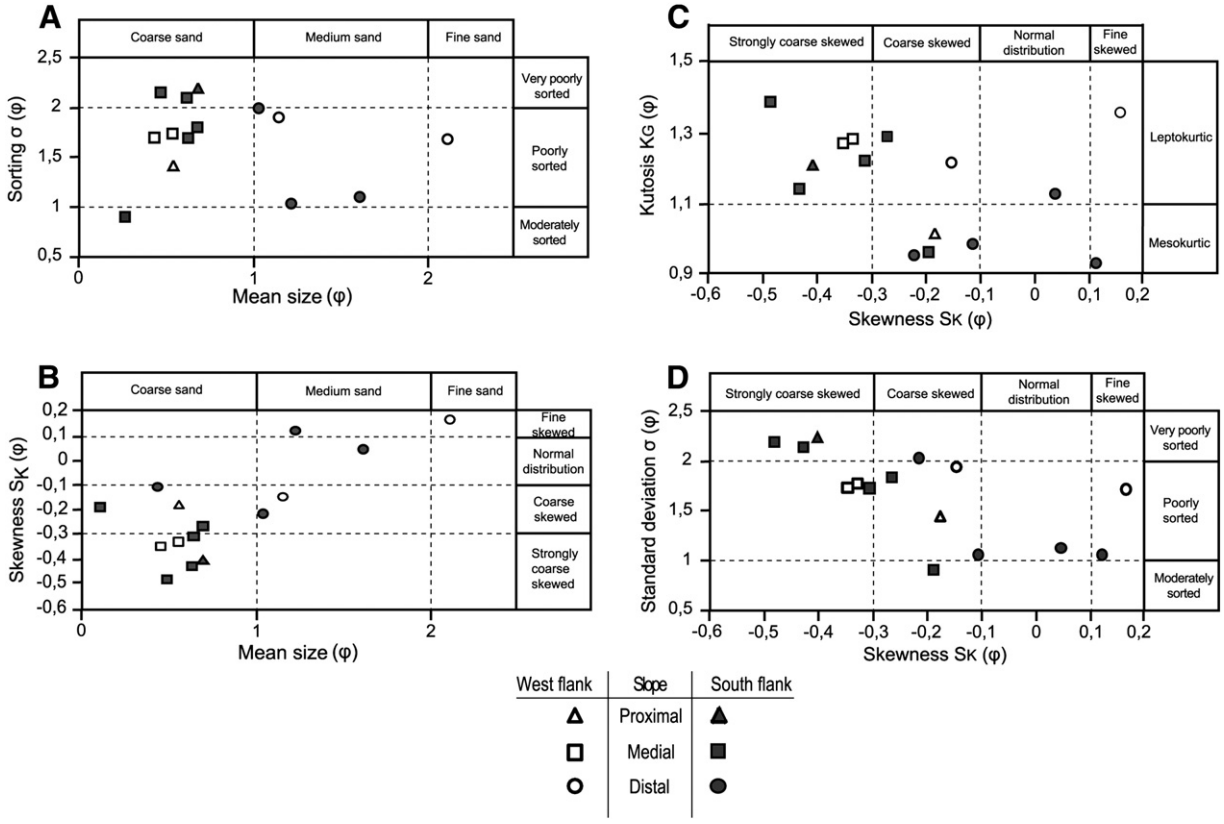


Fig. 10. Grain-size analysis of the matrix of lahar deposits samples ($n = 15$).

possesses a cellphone and can receive text messages sent by watchers. Moreover, social networks like *Facebook* or *Twitter* help to transmit information. (3) Hundreds of people work to quarry volcaniclastic sediments from the bottom of the valleys each day, and at most extraction sites many workers have radios to initiate an evacuation if a lahar is reported.

Some communities have formed associations such as the Union of the Residents of the Southern Boyong River which often gathers academics, engineers and pupils from elementary to high school for discussions about the environmental changes and the risks generated by lahars. Another association, *Pecinta Alam*, decided shortly after the eruption to focus on the potential dangers from lahars. They organized a very effective monitoring system run by local people who provide regular reports on the general state of the Boyong River (color, depth, velocity, etc.). They send these observations to people living near the river, and their data is published online, providing close to real time evaluation of river conditions. Moreover, they prepared an evacuation plan for locals and marked building walls with arrows to indicate the direction to safer ground. Community meetings are also held to show videos of lahars and explain security measures. All these initiatives were taken without government input, as it remains difficult for the political authorities to organize timely evacuations for lahars given the short warning often provided before an event (Major et al., 2003). This community-based risk and crisis management has prevented significant fatalities, and provides an example of a successful method of managing lahar hazard.

4. Discussion

4.1. Rain-triggered lahars following major eruptions of Merapi volcano

Every explosive eruption of Merapi volcano during the 20th century was followed by frequent rain-triggered lahars (Table 4). While the rain-triggered lahars following the 2010 explosive eruption can be compared to the lahars from earlier explosive events in the 20th century, they were much more frequent and generated more extensive damage in just a few months. After the 1930–1931 eruption, lahars were reported on 8 rivers, and after the 1969 eruption, they were reported on 11 rivers. In contrast, lahars following the 1994 dome-collapse PDCs were restricted to the Boyong River (21 occurrences in 1994–1995), and did not extend beyond 13 km from the summit (Lavigne et al., 2000a). After the 2010 eruption, 240 lahars were reported within six months (October 2010 to early May 2011). This almost equals the total number of the lahars reported between 1969 and 1978 (253), and well exceeds the total of 195 lahars reported between 1931 and 1932 (Lavigne et al., 2000a). Frequencies of 2010–2011 rain-triggered lahars are higher than previous lahar crisis over the 20th century. This may be explained by: (1) a lack of information and data for lahars following the 1930–1931 eruption and (2) rainfall intensities. Rainfall data show that the 2010–2011 rainy season reached a total of 17,436 mm of cumulated rainfalls at Merapi, which is higher than the mean rainfall recorded during each rainy season of the 2010 decade, which did not exceed 12,500 mm (De Bélizal, 2012). This important rainfall may be linked to La Niña climatic context which occurred in 2010 (NOAA, 2012). Major lahar crisis at Merapi are thus closely linked to major eruptions which deposit large volume of pyroclastic materials.

Similar lahar crisis following major eruptions can be found on other tropical stratovolcanoes, such as Mount Pinatubo (Luzon, Philippines) or Soufriere Hills (Montserrat, British West Indies). The Plinian explosive eruption of Pinatubo (June 1991) was followed by nearly a decade of lahar activity which killed 600 people and led to the massive relocation of 42,000 families (Tayag and Punongbayan, 1994; Arboleda and Martinez, 1996; Major et al., 1996; Gaillard et al., 2001). As with the 2010 Merapi eruption, frequency of lahars at Pinatubo from 1991 to 1993 was quite high from 0.6 to 1 event per day (Pierson et al., 1996).

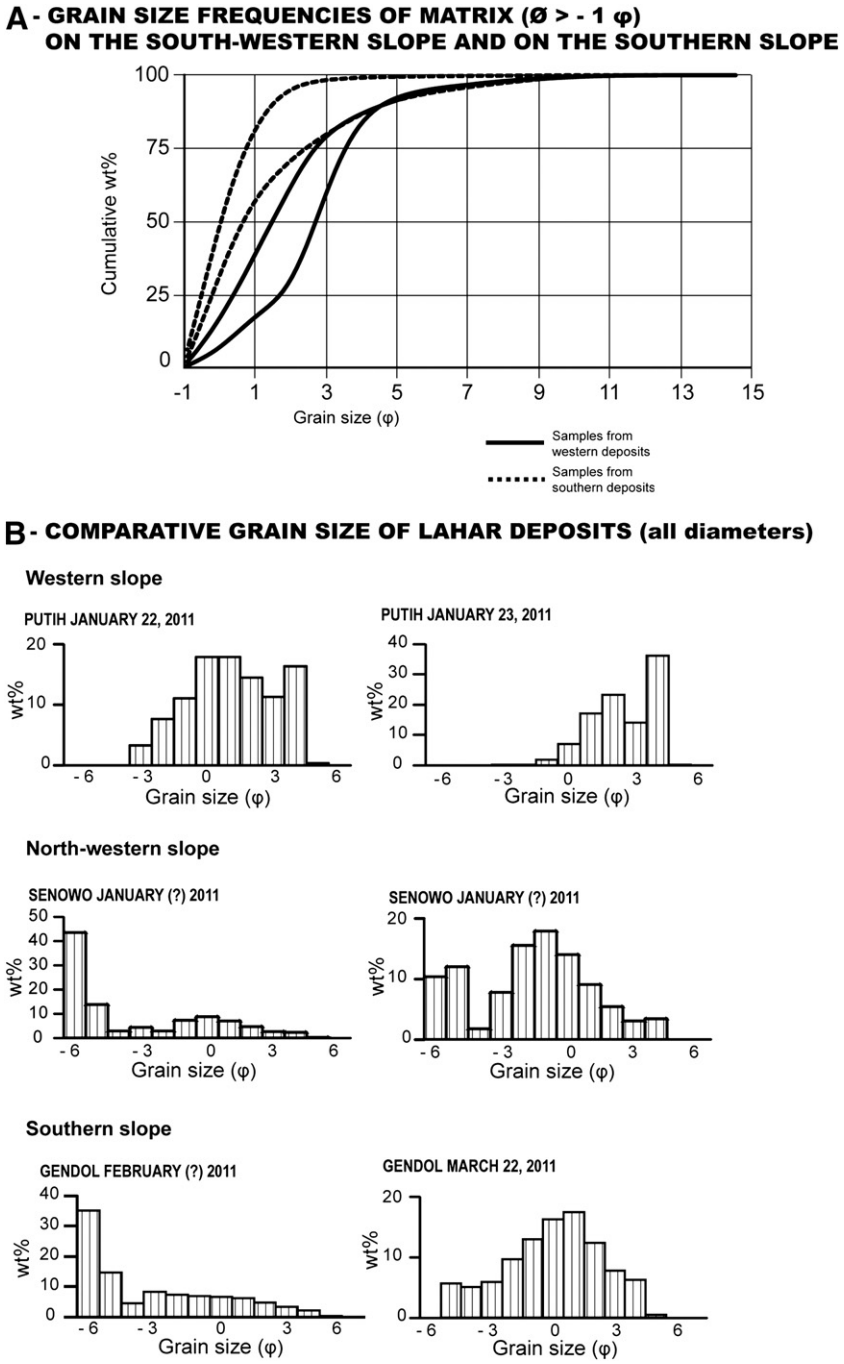


Fig. 11. A: Comparison of composition between the southwestern slope and the southern slope. B: Histograms of grain-size analysis of matrix.

Similarly, in Montserrat, pyroclastic deposits from the 1997 and 2008–2010 eruptions (Pattullo, 2000; Rozdilsky, 2001; OPM, 2010) were remobilized by lahars that extended the length of Belham Valley and formed a volcaniclastic aggradant lobe at the mouth of the Belham. Villages located on the banks of the Belham Valley on the western slope of the volcano are progressively disappearing (Rozdilsky, 2001; OPM, 2010). These examples of post-eruptive rain-triggered lahars show that the impacts of explosive eruptions extend beyond the timeline of the eruptive phases, and can last for many years.

4.2. Perspective: a major risk for the following years?

Damage by the 2010–2011 lahars was particularly important to houses, land and infrastructures at Merapi. Compared to rain-triggered lahars following other Merapi eruptions during the 20th century, 2010–2011 lahars were more numerous and generated more damages than previous lahars following explosive eruptions (Table 4): after the 1969 eruption, lahars destroyed 532 houses and swept away nine bridges within 3 years, which is lower than damage generated by the 2010–2011 lahars in a few months. This is partly due to the changing demography and land use at Merapi volcano, mainly on the ringplain of the volcano. This population is considered very "vulnerable" to natural hazards by many disaster risk researchers (Laksono, 1988; Dove, 2007a,b, 2008; Lavigne et al., 2008). Although the eruption is finished, lahar hazards still threaten communities around the Merapi ringplain, where damage was particularly high (860 houses).

The Indonesian government allocated resources for the reconstruction and rehabilitation of areas affected by lahars. To date, the National

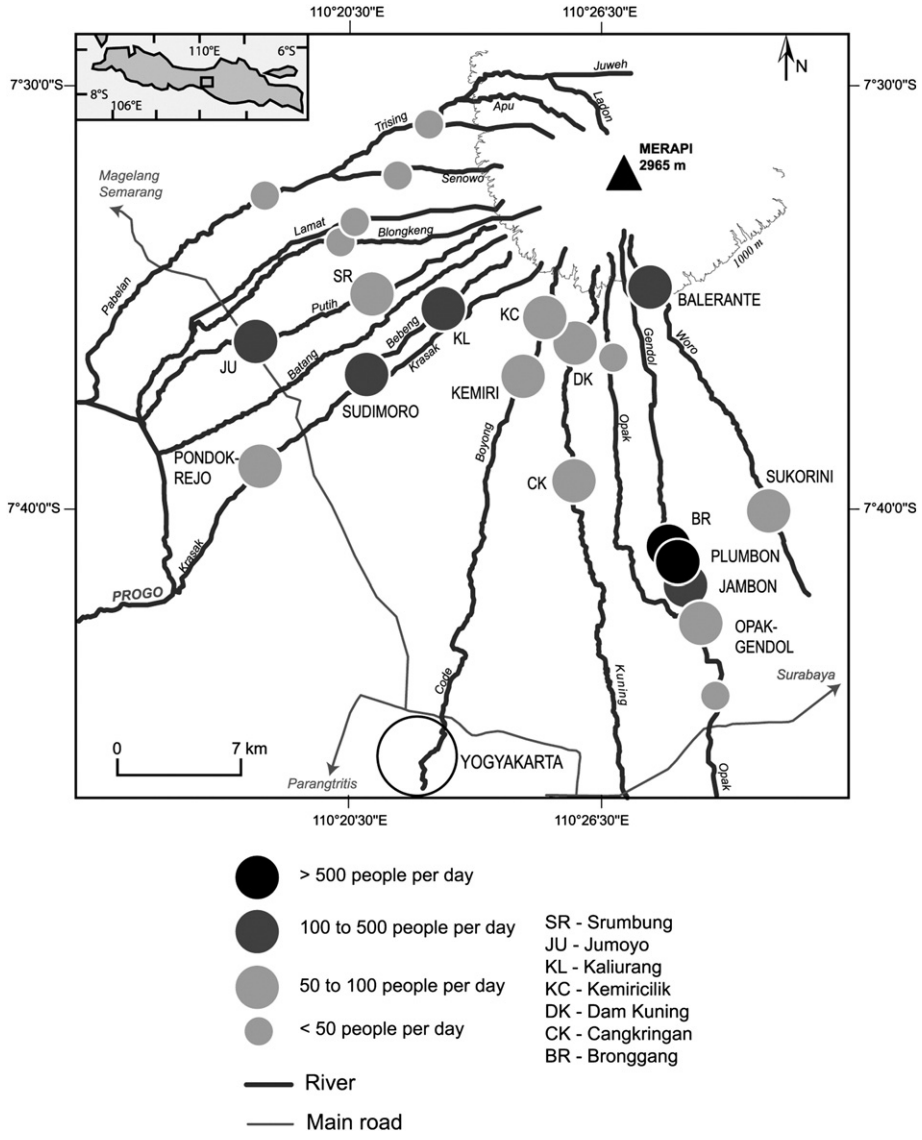


Fig. 12. Location of main quarries on lahar corridors around Merapi volcano.

Office for Disaster Reduction (*Badan Nasional Penanggulangan Bencana*, BNPB) has invested 640,000 USD to build a new bridge on the Yogyakarta–Semarang highway, in order to prevent flooding by lahars from the Putih River. Moreover, 10 bridges have been rebuilt in 2011–2012. The new bridges do not rely on piers positioned along valley bottoms, but instead rely on suspension between consolidated walls on the riverbanks. Although the presence of sabo-dams in upstream location may be dangerous (Lube and Cronin, 2008; Thouret et al., 2010; Lube et al., 2011), a budget of 135 million USD was allocated for repairing and building 77 sabo-dams (sediment dams), which play a major role in controlling the speed, sediment load and discharge of lahars (Lavigne et al., 2000a,b). The government plans to replace and refurbish the water piping in Sleman district for a total of 1.5 million USD, which will facilitate access to water for about 56,000 people in areas where local wells, springs and irrigation channels were destroyed by lahars. The figures quoted here do not account for social aid for housing reconstruction that maybe provided by the government and NGOs (Non Governmental Organizations). However, the Indonesian Ministry of Public Works suggests that a total of 16 million USD has already been invested for lahar recovery at Merapi volcano. This cost will climb if damaging lahars continue to persist in the coming years. At Pinatubo (Philippines), the total loss due to lahars was estimated at nearly one billion dollars about ten years after the 1991 eruption (Gaillard et al., 2001).

Prevention and preparedness should be improved to strengthen the community-based crisis management: scientists and communities have begun to cooperate closely for a better hazard-assessment at a local scale. If rain-triggered lahars following the 1969 Merapi eruption killed 38 people, the 2010–2011 lahars generated 3 fatalities, thanks to an efficient crisis management (see Section 3.4). To improve the hazard assessment and the preparedness of local communities, which is a key factor of risk reduction (De La Cruz-Reyna et al., 2000), the local office at Yogyakarta (BPPTK) of the Center of Volcanological and Geological Hazard Mitigation of Indonesia is publishing a series of hazard maps at 1/12,000 scale centered on the downstream parts of rivers, to help authorities and local populations to adapt. The maps are based on field surveys, remote sensing and flow modeling using a high-resolution DEM and highlight areas where lahars may spill out and generate overflows. Evacuation roads are clearly indicated to help people identify the safest direction in case of emergency. These maps will require regular updating to ensure that hazard plans are as accurate as possible. As villagers need to know that official hazard mapping and evacuation roads have been decided, the maps must be distributed widely and posted in villages where lahars represent a risk. Volunteers already monitoring the rivers should use these plans to improve their early warning

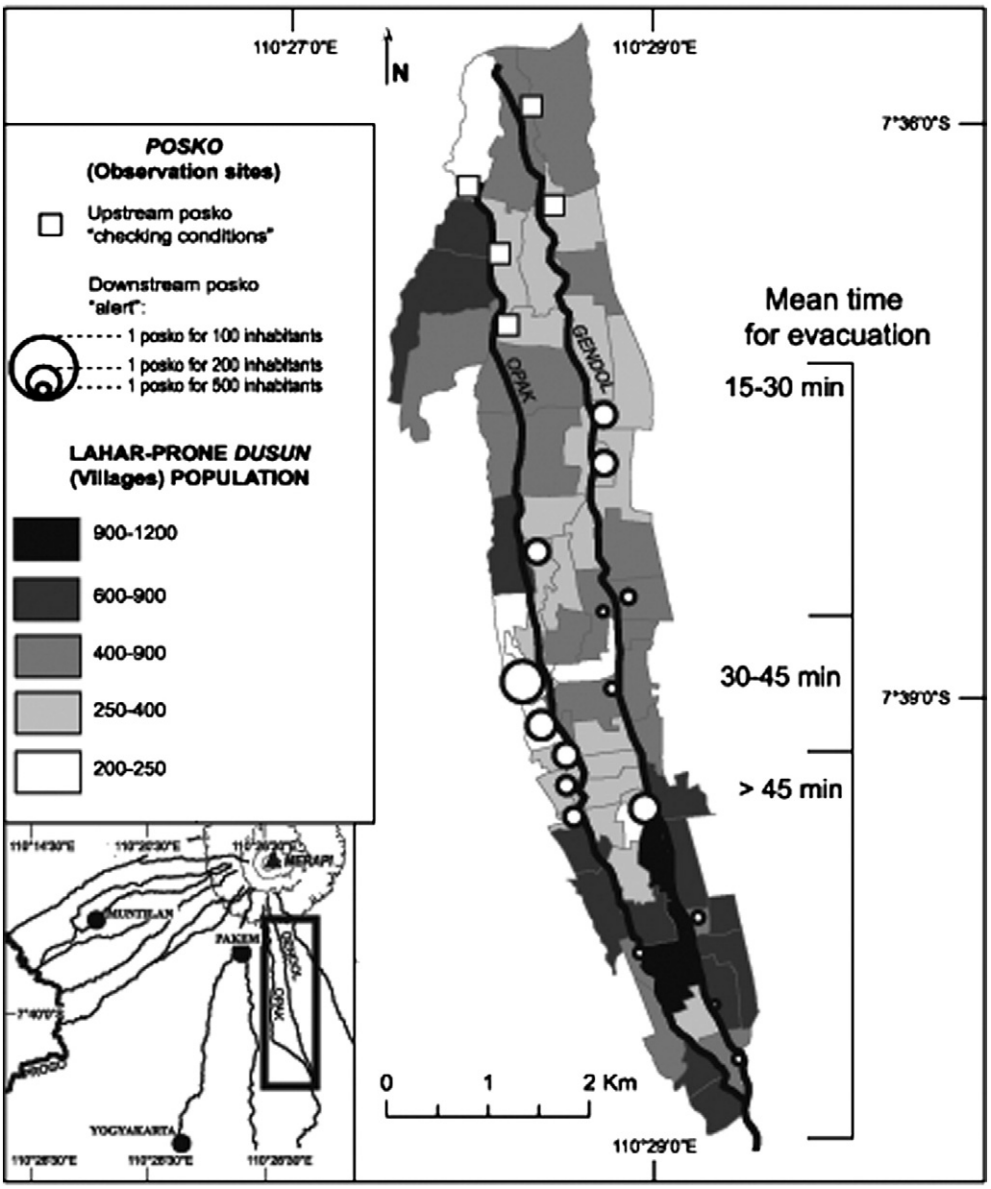


Fig. 13. Community-based look-out stations (poskos) on the Opak and the Gendol Rivers.

systems. If these steps are taken, it may generate a beneficial cooperation between scientists and communities, using reliable scenarios and models. At present, only short-term warning is performed, but these plans will help construct long-range warning and management of lahars. These two mitigation tools thus need to be used simultaneously, as seen at Pinatubo volcano during the 1990s (Janda et al., 1996), to construct a more effective hazard management plan.

5. Conclusion

We proposed in this paper a comprehensive approach which aimed to put together all the different components of lahar-related issues at Merapi volcano in order to understand why lahars represent a major risk after the 2010 VEI 4 eruption. We showed that:

(1) Lahars were frequent and occurred on almost all the watersheds under the active cone of the volcano. They were widespread and extended more than 15 km from the crater.
(2) Lahars rapidly formed large corridors on the distal slope of the volcano. Avulsions, riverbank erosion and riverbed downcutting were presented at a local scale, and the related damages were exposed. Lahars totally buried 215 houses and damaged 645 houses, and led to major disturbances in traffic (destroyed bridges and roads).
(3) Volcaniclastic sediments brought by 2010–2011 lahars extend beyond 20 km on the distal slope. Description of lithofacies and grain-size analysis showed that most lahars in distal areas were dilute and carried few boulders. Large-magnitude events can reach the distal part of the Pabelan River. Deposits are mainly supported by a coarse sand matrix, which represents a valuable economic resource, attracting people to lahar corridors to quarry the deposits and exposing them to a frequent hazard.
(4) An efficient community-based hazard management prevented significant human losses. Distribution of information on headwaters conditions of the rivers by networks of river watchers allowed quick evacuation of exposed people at least 30 min before the lahars reach their locations.

Contrary to what was shown in previous overviews concerning lahars at Merapi volcano following dome-collapse PDCs (Lavigne et al., 2000a,b), 2010–2011 rain-triggered lahars following a VEI 4 eruption

Table 4
Eruptions and rain-triggered lahars at Merapi volcano since the 20th century.
From Lavigne et al. (2000a,b), Lavigne and Thouret (2002), Voight et al. (2000) and Charbonnier and Gertisser (2008).

Eruptions					Rain-triggered lahars					
Year	Volcanic explosivity index (VEI)	Pyroclastic volume	Max. runout distance of PDCs	Direction of PDCs	Year	Number of events	Affected rivers	Max runout distance	Magnitude	Damages
		($m^3 \times 10^6$)	(km)					(km)		
1920	2	5	5.5	W	?	?	?	?	?	1 village 35 fatalities
1930	3	26	13	S–SW	1930–1931	33	Ba	>15		4 bridges 340 ha of farmlands
					1931–1932	162	Se, Pa, La, Bl, Ba, Be, Kr, Bo, Ku			1 village 2 bridges
1934	2	?	7	W						
1939	2	?	3	S–SW						
1942	2	4	2.5	SW						
1948	2	?	?	SW						
1953	2	20	7	W–NW	1954	1	Pa	?	Max. depth 2 m	
1957	1	8.5	6	W		?				
1961	3	29.4	12	SW	1961–1963	?	Se, Pa, Bl, Ba		Max. depth>5 m	95 houses 7 fatalities 1 bridge
1969	2	12.6	13.3	SW	1969	44	Se, Pa, La, Bl,		Max. depth>5 m	751 houses 13 bridges 370 ha of farmlands
					1970–1971	21	Se, Pa, La, Ba, Be, Kr, Bo, Ku, Op, Ge, Wo			
1972	2	6.5	7	SW	1972–1973	17	Pu, Be, Kr, Bo, Ku		Max. depth>5 m	75 houses
					1974	21				134 houses 1 bridge 30 ha of farmlands
					1975	65				
1976	2	1.2	3.5	SW	1976–1978	85	Bl, Pu, Ba, Be, Kr, Bo, Ku		Max. depth>5 m	323 houses 3 bridge 330 ha of farmlands
1979	2	?	?	SW						
1984	2	7	7	SW	1985–1990	27	Pu		Max. peak discharge 2000 $m^3\ s^{-1}$	
1992	2	3	4	SW	1992–1993	9	Pu			
1994	2	2.5	6	S	1994–1996	42	Bo	13	Max. peak discharge 360 $m^3\ s^{-1}$	27 trucks swept by lahars 1 bridge
2001	?	?	?	SW						
2006	?	13.3	7	SE	2006–2009	?	Ge	<8		
2010	4	30–60	17	NW–W–SW–S–SE	Oct. 2010 to Jan. 2012	282	Ld, Jw, Ap, Tr, Se, Pa, La, Bl, Pu, Ba, Be, Kr, Bo, Ku, Op, Ge, Wo.	>20	Max. peak discharge 1800 $m^3\ s^{-1}$	860 houses 21 bridges 3000 affected people 3 fatalities

Ld: Ladon; Jw: Juweh; Ap: Apu; Tr: Trising; Se: Senowo; Pa: Pabelan; La: Lamat; Bl: Blongkeng; Pu: Putih; Ba: Batang; Be: Bebeng; Kr: Krasak; Bo: Boyong; Ku: Kuning; Op: Opak; Ge: Gendol; and Wo: Woro.

do not only extend on the medial slope of the volcano, but reached locations > 15 km from the crater, where they form large corridors. This shift in the hazard-prone areas at Merapi needs further assessment, and further studies should focus on the distal slope of the volcano where lahars will represent a risk in the coming years.

Acknowledgment

This study was undertaken within the framework of the Mitigate and Assess Risk from Volcanic Impact on Terrain and Human Activities MIAVITA, under Work Package 5: "Socio-economic Vulnerability and Resilience". The MIAVITA project is financed by the European Commission under the 7th Framework Programme for Research and Technological Development, Area "Environment", Activity 6.1 "Climate Change, Pollution and Risks". The article reflects the authors' views. The European Commission is not liable for any use that may be made of the information contained therein. Additional funding was granted by the "Beasiswa Unggulan" from the Indonesian Ministry of National Education (*Perencanaan dan Kerjasama Luar Negeri*, *Kementerian Pendidikan dan Kebudayaan Republik Indonesia*).

The authors wish to thank Dr. Surono (Director of the Centre for Volcanology and Geological Hazard Mitigation, Indonesia) and Dr. Subandriyo, Head of the Merapi Volcano Observatory. We also acknowledge all the volunteers who took time to explain their work. Special thanks to Tawia Abbam (University of Southampton) for the English editing.

We are grateful to J.J. Major, C. Gomez and one anonymous reviewer who provided in-depth reviews and valuable advices which improved the manuscript. Guest editors P. Jousset and J. Pallister are also to be thanked for their patience and helpful comments.

References

Abdurachman, E.K., Bourdier, J.-L., Voight, B., 2000. Nuées ardentes of 22 November 1994 at Merapi volcano, Central Java, Indonesia. Journal of Volcanology and Geothermal Research 100, 345–361.

Andreastuti, S.D., Alloway, B.V., Smith, I.E.M., 2000. A detailed tephrostratigraphic framework at Merapi Volcano, Central Java, Indonesia: implications for eruption prediction and hazard assessment. Journal of Volcanology and Geothermal Research 100, 51–67.

Arboleda, A., Martinez, Ma.M.L., 1996. 1992 lahars in the Pasig–Potrero System. In: Newhall, C.G., Punongbayan, R.S. (Eds.), Fire and Mud — Eruptions and Lahars of Mount Pinatubo, Philippines. Philippine Institute of Volcanology and Seismology and the University of Washington Press (1126 pp.).

Bautista, C.B., 1996. The Mount Pinatubo disaster and the people of Central Luzon. In: Newhall, C.G., Punongbayan, R.S. (Eds.), Fire and Mud: eruptions and lahars of Mount Pinatubo, Philippines. Phivolcs/University of Washington Press. 1126 pp.

Capra, L., Poblete, M.A., Alvarado, R., 2004. The 1997 and 2001 lahars of Popocatépetl volcano (Central Mexico): textural and sedimentological constraints on their origin and hazards. Journal of Volcanology and Geothermal Research 131, 351–369.

Charbonnier, S., Gertisser, R., 2008. Field observations and surface characteristics of pristine block-and-ash flow deposits from the 2006 eruption of Merapi Volcano, Java, Indonesia. Journal of Volcanology and Geothermal Research 177, 971–982.

Charbonnier, S., Germa, A., Connor, C.B., Gertisser, R., Preece, K., Komorowski, J.C., Lavigne, F., Dixon, T., Connor, L., 2013. Evaluation of the impacts of the 2010 pyroclastic density currents at Merapi volcano from high-resolution satellite imagery, field investigations and numerical simulations. Journal of Volcanology and Geothermal Research 261, 295–315.

Cronin, S.J., Neall, V.E., Lecointre, J.A., Palmer, A.S., 1997. Changes in Whangaehu river lahar characteristics during the 1995 eruption sequence, Ruapehu volcano, New Zealand. Journal of Volcanology and Geothermal Research 76, 47–61.

De Bélizal, E., 2012. Les corridors de lahars, des espaces entre risque et ressource. PhD dissertation, Paris 1 Pantheon-Sorbonne University, 413 pp.

De Bélizal, E., Lavigne, F., Grancher, D., 2011. Quand l'aléa devient la ressource: l'activité d'extraction des matériaux volcaniques autour du volcan Merapi (Indonésie) dans la compréhension des risques locaux. Cybergeo: European Journal of Geography [On line], Environnement, Nature, Paysage. URL: http://cybergeo.revues.org/23555.

De La Cruz-Reyna, S., Meli, R.P., Quaas, R.W., 2000. Volcanic crisis management. In: Sigurdsson, H., Houghton, B.F., McNutt, S.R., Rymer, H., Stix, J. (Eds.), Encyclopedia of Volcanoes. Academic Press, San Diego, pp. 1199–1214.

Dove, M.R., 2007a. Volcanic eruption as metaphor of social integration: a political ecological study of Mount Merapi, Central Java. In: Connell, J., Waddell, E. (Eds.), Environment, Development and Change in Rural Asia–Pacific: Between Local and Global. Routledge, Londres/NY, pp. 16–37.

Dove, M.R., 2007b. Perceptions of local knowledge and adaptation on Mt. Merapi, Central Java. In: Ellen, R.F. (Ed.), Modern Crises and Traditional Strategies: Local Ecological Knowledge in Island Southeast Asia. Berghahn Books, New York/Oxford, pp. 238–262.

Dove, M.R., 2008. Perception of volcanic eruption as agent of change on Merapi volcano, Central Java. Journal of Volcanology and Geothermal Research 172, 329–337.

Folk, R.L., Ward, W.C., 1957. Brazos River bar: a study in the significance of grain size parameters. Journal of Sedimentary Petrology 27, 3–26.

Gaillard, J.C., D'Ercole, R., Leone, F., 2001. Cartography of population vulnerability to volcanic hazards and lahars of Mount Pinatubo (Philippines): a case study in Pasig–Protero River basin (province of Pampanga). Géomorphologie: relief, processus, environnement 3, 209–222.

Inman, D.L., 1952. Measures for describing the size distribution of sediments. Journal of Sedimentary Research 22, 125–145.

Janda, R.J., Daag, A.S., Delos Reyes, P.J., Newhall, C.G., Pierson, T.C., Punongbayan, R.S., Rodolfo, K.S., Solidum, R.U., Umbal, J.V., 1996. Assessment and response to lahar hazard around Mount Pinatubo, 1991 to 1993. In: Newhall, C.G., Punongbayan, Raymundo S. (Eds.), Fire and Mud — Eruptions and Lahars of Mount Pinatubo, Philippines. Philippine Institute of Volcanology and Seismology and the University of Washington Press (1126 pp.).

Jitousono, T., Shimokawa, E., Tsuchiya, S., Haryanto, Djamal H., 1995. Debris flow following the 1984 eruption with pyroclastic flows in Merapi volcano, Indonesia. Final Proc. Workshop on Erosion Control through Volcanic Hydrological Approach (WECVHA), Sabo Technical Center, Yogyakarta, Indonesia, pp. 131–149.

Komorowski, J.C., Jenkins, S., Baxter, P., Picquout, A., Lavigne, F., Charbonnier, S., Gertisser, R., Preece, K., Cholik, N., Budi-Santoso, A., Surono, 2013. Paroxysmal dome explosion during the Merapi 2010 eruption: processes and facies relationships of associated high-energy pyroclastic density currents. Journal of Volcanology and Geothermal Research 261, 260–294.

Laksono, P.M., 1988. Perception of volcanic hazards: villagers versus government officials in Central-Java. In: Dove, M.R. (Ed.), The real and Imagined Role of Culture in Development: Case Studies from Indonesia. University of Hawaii Press, Honolulu (289 pp.).

Lavigne, F., 1999. Lahars hazard micro-zonation and risk assessment in Yogyakarta city, Indonesia. GeoJournal 49, 131–138.

Lavigne, F., Thouret, J.-C., 2002. Sediment transportation and deposition by rain-triggered lahars at Merapi volcano, Central Java, Indonesia. Geomorphology 49, 45–69.

Lavigne, F., Thouret, J.C., Suwa, H., Voight, B., Young, K., Lahusen, R., Marso, J., Sumaryono, A., Sayudi, D.S., Dejean, M., 2000a. Instrumental lahar monitoring at Merapi Volcano, Central Java, Indonesia. Journal of Volcanology and Geothermal Research 100, 457–478.

Lavigne, F., Thouret, J.C., Voight, B., Suwa, H., Sumaryono, A., 2000b. Lahars at Merapi volcano: an overview. Journal of Volcanology and Geothermal Research 100, 423–456.

Lavigne, F., De Coster, B., Juvin, N., Flohic, F., Gaillard, J.C., Texier, T., Morin, J., Sartohadi, J., 2008. People's behaviour in the face of volcanic hazards: perspectives from Javanese communities, Indonesia. Journal of Volcanology and Geothermal Research 172, 273–287.

Lube, G., Cronin, S.J., 2008. 2008: the avulsion of pyroclastic flows from channels — a cautionary tale from the 14 June 2006 eruption of Gunung Merapi, Indonesia. Asian International Symposium on Modelling of Volcanic Eruption for Volcanic Hazard Assessment, Bandung, Indonesia, 1–4 December 2008. Centre for Volcanology and Geological Hazard Mitigation, Bandung, Indonesia, pp. 67–71. in Extended Abstracts.

Lube, G., Cronin, J.S., Thouret, J.-C., Surono, 2011. Kinematic characteristics of pyroclastic density currents at Merapi and controls on their avulsion from natural and engineered channels. Geological Society of America Bulletin 123, 1127–1140.

Major, J.J., Janda, R.J., Daag, A.S., 1996. Watershed disturbance and lahars on the east side of Mount Pinatubo during the mid-June 1991 eruptions. In: Newhall, C.G., Punongbayan, R.S. (Eds.), Fire and Mud — Eruptions and Lahars of Mount Pinatubo, Philippines. Philippine Institute of Volcanology and Seismology and the University of Washington Press (1126 pp.).

Major, J.J., Schilling, S.P., Pullinger, C.R., 2003. Volcanic debris flows in developing countries—the extreme need for public education and awareness of debris-flow hazards. In: Rickenmann, D., Chen, C.L. (Eds.), Debris-Flow Hazards Mitigation—Mechanics, Prediction, and Assessment, Proceedings of the Third International Conference, Davos, Switzerland, September 10–12, Rotterdam. MillPress Science Publishers, pp. 1185–1196.

Mei, E.T.W., Lavigne, F., in press. Evacuation Management during the 2010 Eruptive Crisis of Merapi Volcano, Journal of International Emergency Management.

Newhall, C.G., Bronto, S., Alloway, B., Banksa, N.G., Bahar, I., Del Marmol, M.A., Hadisantono, R.D., Holcomb, R.T., McGeehin, J., Miksic, J.N., Rubin, M., Sayudi, S.D., Sukhyar, R., Andreastuti, S., Tilling, R.I., Torley, F., Trimble, D., Wirakusumah, A.D., 2000. 10,000 Years of explosive eruptions of Merapi Volcano, Central Java: archaeological and modern implications. Journal of Volcanology and Geothermal Research 100, 9–50.

National Oceanic and Atmospheric Administration (NOAA), 2012. Cold and warm episodes by season. Changes to the Oceanic Nino Index. NOAA website: http://www.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ensoyears.shtml.

Pattullo, P., 2000. Fire from the Mountain: The Tragedy of Montserrat and the Betrayal of Its People.Constable and Robinson, London 218 pp.

Pierson, T.C., 1995. Flow characteristics of large eruption-triggered debris flows at snow-clad volcanoes: constraints for debris-flow models. Journal of Volcanology and Geothermal Research 66, 283–294.

Pierson, T.C., Scott, K.M., 1985. Downstream dilution of a lahar: transition from debris flow to hyperconcentrated streamflow. Water Resources Research 21, 1511–1524.

Pierson, T.C., Daag, A.S., Delos Reyes, P.J., Regalado, Ma.T.M., Solidum, R.U., Tubianosa, B.S., 1996. Flow and deposition of posteruption hot lahars on the east side of Mount Pinatubo, July–October 1991. In: Newhall, C.G., Punongbayan, R.S. (Eds.), Fire and Mud — Eruptions and Lahars of Mount Pinatubo, Philippines. Philippine Institute of Volcanology and Seismology and the University of Washington Press (1126 pp.).

Rozdilsky, J.L., 2001. Second hazards assessment and sustainable hazards mitigation: disaster recovery on Montserrat. Natural Hazards Review 2, 64–71.

Schwarzkopf, L.M., Schminke, H.U., Cronin, S.J., 2005. A conceptual model for block-and-ash flow basal avalanche transport and deposition based on deposits architecture of 1998 and 1994 Merapi flows. Journal of Volcanology and Geothermal Research 139, 117–134.

Scott, K.M., Vallance, J.W., Pringle, P.T., 1995. Sedimentology, behavior, and hazards of debris flows at Mount Rainier, Washington. U.S. Geological Survey Professional Paper, 1547 (56 pp.).

Scott, K.M., Janda, R.J., De la Cruz, E.G., Gabinete, E., Eto, I., Isada, M., Sexon, M., Hadley, K.C., 1996. Channel and sedimentation responses to large volumes of 1991 volcanic deposits on the east flank of Mount Pinatubo. In: Newhall, C.G., Punongbayan, R.S. (Eds.), Fire and Mud — Eruptions and Lahars of Mount Pinatubo, Philippines. Philippine Institute of Volcanology and Seismology and the University of Washington Press (1126 pp.).

Smith, G.A., Fritz, W.J., 1989. Penrose Conference report: volcanic influences on terrestrial sedimentation. Geology 17, 376.

Surono, Jousset, P., Pallister, J., Boichu, M., Buongiorno, M.F., Budisantoso, A., Costa, F., Andreastuti, S., Prata, F., Schneider, D., Clarisse, L., Humaida, H., Sumarti, S., Bignami, C., Griswold, J., Carn, S., Oppenheimer, C., Lavigne, F., 2012. The 2010 explosive eruption of Java's Merapi volcano — a '100-year' event. Journal of Volcanology and Geothermal Research 241–242, 121–135.

Tayag, J.C., Punongbayan, R.S., 1994. Volcanic disaster mitigation in the Philippines: experience from Mt. Pinatubo. Disasters 18, 1–15.

Thierry, P., Jousset, P., Boichu, M., Surono, Buongiorno F., Bignami, C., Lavigne, F., Picquout, A., Le Cozanet, G., Jenkins, S., Berger, J., Mei, E., 2011. Merapi volcano 2010 crisis: a case study for implementing the FP7 MIAVITA European project, European Geoscience Union General Assembly. European Geoscience Union General Assembly, Vienna, Austria.

Thouret, J.-C., Lavigne, F., Kelfoun, K., Bronto, S., 2000. Toward a revised hazard assessment at Merapi volcano, Central Java. Journal of Volcanology and Geothermal Research 100, 479–502.

Thouret, J.C., Lube, G., Liew, S.C., Cronin, S.J., Surono, 2010. Analysis of the 2006 eruption deposits of Merapi Volcano, Java, Indonesia, using high-resolution IKONOS images and complementary ground based observations. Remote Sensing of Environment 114, 1949–1967.

Vallance, J.W., 2000. Lahars. In: Sigurdsson, H., Houghton, B., McNutt, S.R., Rymer, H., Stix, J. (Eds.), Encylopedia of volcanoes. Academic Press, San Diego (1417 pp.).

Voight, B., Constantine, E.K., Siswowidjoyo, S., Torley, R., 2000. Historical eruptions of Merapi Volcano, Central Java, Indonesia, 1768–1998. Journal of Volcanology and Geothermal Research 100, 69–138.